



Report No: PADHI00264

INTERNATIONAL DEVELOPMENT ASSOCIATION

PROJECT APPRAISAL DOCUMENT
ON A
PROPOSED CREDIT
IN THE AMOUNT OF US\$112.55 MILLION

AND A
PROPOSED SHORTER MATURITY LOAN
IN THE AMOUNT OF US\$8.45 MILLION

AND A TRUST FUND GRANT
IN THE AMOUNT OF
US\$9.0 MILLION
FROM THE KYRGYZ REPUBLIC – WATER SUPPLY AND SANITATION UNIVERSAL ACCESS PROGRAM
MULTI-DONOR TRUST FUND

TO THE

KYRGYZ REPUBLIC

FOR A

WATER SUPPLY AND SANITATION UNIVERSAL ACCESS PROGRAM – 1 PROJECT
(P500620)
PHASE 1 OF THE MULTI-PHASE PROGRAMMATIC APPROACH

WITH AN OVERALL FINANCING ENVELOPE OF US\$200 MILLION
For Phase 1:
March 27, 2025

Water
Europe And Central Asia

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CURRENCY EQUIVALENTS

(Exchange Rate Effective February 28, 2025)

Currency Unit = KYRGYZ SOM (KGS)

KGS 87.48 = US\$1.00

FISCAL YEAR

January 1 - December 31

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ABBREVIATIONS AND ACRONYMS

ADB	Asian Development Bank
AIIB	Asian Infrastructure Investment Bank
AO	Aiyl Okmotu
ARIS	Community Development and Investment Agency
CDWUU	Community Drinking Water User Union
CPF	Country Partnership Framework
CRWSP	Climate Resilient Water Services Project
DA	Designated Account
DFIL	Disbursement and Financial Information Letter
DWSSP	District Water and Sanitation Service Provider
E&S	Environmental and Social
ERR	Economic Rate of Return
ESCP	Environmental and Social Commitment Plan
ESMF	Environmental and Social Management Framework
ESSs	Environmental and Social Standards
FM	Financial Management
FSM	Fecal Sludge Management
GCP	Global Challenge Program
GHG	Greenhouse Gas Emissions
GoKR	Government of the Kyrgyz Republic
GRS	Grievance Redress Service
IA	Implementing Agency
IDA	International Development Association
IPF	Investment Project Financing
IsDB	Islamic Development Bank
LMP	Labor Management Procedures
M&E	Monitoring and Evaluation
MFD	Maximizing Finance for Development
MFD-EP	MFD Enabling Project
MoF	Ministry of Finance
MPA	Multiphase Programmatic Approach
MWE	Municipal Water Enterprise
MWRAP	Ministry of Water Resources, Agriculture and Processing Industry
NPV	Net Present Value
NRW	Non-Revenue Water
O&M	Operation and Maintenance
PBG	Performance-Based Grant
PCE	Private Capital Enabling
PCU	Project Coordination Unit
PDDWSSPL	Program for Development of Drinking Water Supply and Sanitation Systems for Populated Localities
PDO	Project Development Objective

PIR	Policies, Institutions and Regulations
PIU	Project Implementation Unit
POM	Project Operations Manual
PP	Procurement Plan
PPL	Public Procurement Law
PPSD	Project Procurement Strategy for Development
PrDO	Program Development Objective
PVDP	Professional and Vocational Development Program
RPF	Resettlement Policy Framework
RWSSP	Rural Water Supply and Sanitation Project
SDC	Swiss Agency for Development and Cooperation
SDG	Sustainable Development Goal
SEP	Stakeholder Engagement Plan
SIASAR	Rural Water Supply and Sanitation Information System
SIDWSWD	State Institution of Drinking Water Supply and Wastewater Disposal
SRWSSDP	Sustainable Rural Water Supply and Sanitation Development Project
STEP	Systematic Tracking of Exchanges in Procurement
TA	Technical Assistance
TOR	Terms of Reference
WASH	Water, Sanitation, and Hygiene
WB-RWIDB	World Bank Resilient Water Infrastructure Design Brief
WRS	Water Resources Service
WSS	Water Supply and Sanitation



TABLE OF CONTENTS

DATASHEET	ii
I. STRATEGIC CONTEXT	1
A. Country Context	1
B. Sectoral and Institutional Context	2
C. Relevance to Higher Level Objectives	5
D. Multiphase Programmatic Approach.....	6
II. PROJECT DESCRIPTION.....	10
A. Project Development Objective	10
B. Project Components	11
C. Project Beneficiaries	14
D. Rationale for Bank Involvement and Role of Partners	14
E. Lessons Learned	15
III. IMPLEMENTATION ARRANGEMENTS	16
A. Institutional and Implementation Arrangements.....	16
B. Results Monitoring and Evaluation Arrangements	17
C. Sustainability	17
IV. PROJECT APPRAISAL SUMMARY	17
A. Technical, Economic, and Financial Analysis	17
B. Fiduciary	19
C. Legal Operational Policies	20
D. Environmental and Social	20
E. Gender, Citizen Engagement, and Climate Change	22
V. GRIEVANCE REDRESS SERVICES	23
VI. KEY RISKS.....	23
VIII. RESULTS FRAMEWORK AND MONITORING	25
ANNEX 1: Implementation Arrangements and Support Plan	34
ANNEX 2: Detailed Project Design, Economic and Financial Analysis, and Project Map.....	41

**DATASHEET****BASIC INFORMATION**

Project Beneficiary(ies)	Operation Name		
Kyrgyz Republic	Kyrgyz Republic: Water Supply and Sanitation Universal Access Program-1 Project		
Operation ID	Financing Instrument	Environmental and Social Risk Classification	
P500620	Investment Project Financing (IPF)	Substantial	

Financing & Implementation Modalities

☒ Multiphase Programmatic Approach (MPA)	☒ Contingent Emergency Response Component (CERC)
☐ Series of Projects (SOP)	☐ Fragile State(s)
☐ Performance-Based Conditions (PBCs)	☐ Small State(s)
☐ Financial Intermediaries (FI)	☐ Fragile within a non-fragile Country
☐ Project-Based Guarantee	☐ Conflict
☐ Deferred Drawdown	☐ Responding to Natural or Man-made Disaster
☐ Alternative Procurement Arrangements (APA)	☐ Hands-on Expanded Implementation Support (HEIS)

Expected Approval Date	Expected Closing Date	Expected Program Closing Date
18-Apr-2025	31-Dec-2029	30-Jun-2035
Bank/IFC Collaboration	Joint Level	
Yes	Complementary or Interdependent project requiring active coordination	

MPA Program Development Objective

To provide universal access and improve the sustainability of water supply and sanitation services in the Kyrgyz Republic.

MPA FINANCING DATA (US\$, Millions)



MPA Program Financing Envelope	400.00
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Components

Component Name	Cost (US\$)
Water Supply Investments	140,000,000.00
Sanitation Development	29,000,000.00
Performance-based Service Improvement Program	22,000,000.00
MPA Program Structuring and Management, and Institutional Development Support	9,000,000.00
Contingent Emergency Response	0.00

Organizations

Borrower:	Kyrgyz Republic
Implementing Agency:	State Institution of Drinking Water Supply and Wastewater Disposal under the Water Resources Service under the Ministry of Water Resources, Agriculture and Processing Industry

MPA FINANCING DETAILS (US\$, Millions)

MPA Program Financing Envelope:	400.00
of which Bank Financing (IBRD):	0.00
of which Bank Financing (IDA):	200.00
of which Other Financing sources:	200.00

PROJECT FINANCING DATA (US\$, Millions)**Maximizing Finance for Development**

Is this an MFD-Enabling Project (MFD-EP)?	Yes
Is this project Private Capital Enabling (PCE)?	Yes

SUMMARY

Total Operation Cost	200.00
Total Financing	200.00



of which IBRD/IDA	121.00
Financing Gap	0.00

DETAILS

World Bank Group Financing

International Development Association (IDA)	121.00
of which IDA Recommitted	3.55
IDA Credit	112.55
IDA Shorter Maturity Loan (SML)	8.45

Non-World Bank Group Financing

Other Sources	70.00
Asian Infrastructure Investment Bank	50.00
OPEC FUND	20.00
Trust Funds	9.00
Trust Funds	9.00

IDA Resources (US\$, Millions)

	Credit Amount	Grant Amount	SML Amount	Guarantee Amount	Total Amount
National Performance-Based Allocations (PBA)	112.55	0.00	8.45	0.00	121.00
Total	112.55	0.00	8.45	0.00	121.00

Expected Disbursements (US\$, Millions)

WB Fiscal Year	2025	2026	2027	2028	2029	2030
Annual	0.00	0.50	19.00	26.00	35.00	40.50



Cumulative	0.00	0.50	19.50	45.50	80.50	121.00
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PRACTICE AREA(S)

Practice Area (Lead)

Water

Contributing Practice Areas

CLIMATE

Climate Change and Disaster Screening

Yes, it has been screened and the results are discussed in the Operation Document

SYSTEMATIC OPERATIONS RISK- RATING TOOL (SORT)

Risk Category

Rating

1. Political and Governance	● Substantial
2. Macroeconomic	● Moderate
3. Sector Strategies and Policies	● Substantial
4. Technical Design of Project or Program	● Moderate
5. Institutional Capacity for Implementation and Sustainability	● Substantial
6. Fiduciary	● Substantial
7. Environment and Social	● Substantial
8. Stakeholders	● Moderate
9. Overall	● Substantial
Overall MPA Program Risk	● Substantial

POLICY COMPLIANCE

Policy

Does the project depart from the CPF in content or in other significant respects?



☐ Yes ☒ No

Does the project require any waivers of Bank policies?

☐ Yes ☒ No

ENVIRONMENTAL AND SOCIAL

Environmental and Social Standards Relevance Given its Context at the Time of Appraisal

E & S Standards	Relevance
ESS 1: Assessment and Management of Environmental and Social Risks and Impacts	Relevant
ESS 10: Stakeholder Engagement and Information Disclosure	Relevant
ESS 2: Labor and Working Conditions	Relevant
ESS 3: Resource Efficiency and Pollution Prevention and Management	Relevant
ESS 4: Community Health and Safety	Relevant
ESS 5: Land Acquisition, Restrictions on Land Use and Involuntary Resettlement	Relevant
ESS 6: Biodiversity Conservation and Sustainable Management of Living Natural Resources	Relevant
ESS 7: Indigenous Peoples/Sub-Saharan African Historically Underserved Traditional Local Communities	Not Currently Relevant
ESS 8: Cultural Heritage	Relevant
ESS 9: Financial Intermediaries	Not Currently Relevant

NOTE: For further information regarding the World Bank's due diligence assessment of the Project's potential environmental and social risks and impacts, please refer to the Project's Appraisal Environmental and Social Review Summary (ESRS).

LEGAL

Legal Covenants

Sections and Description

Financing Agreement: Schedule 2, Section I.A.1(a). The Recipient, through the SIDWSWD, shall at the central level, by no later than thirty (30) days after the Effective Date, (i) establish a Project Coordination Unit ("PCU") to be responsible for performing key management functions of the Project, including coordination, financial management, procurement, monitoring and evaluation, social and environmental standards management, verification, and reporting functions, with mandates, functions, responsibilities, structures, resources and staff, as further described in the POM and satisfactory to the Association; (ii) hire a disbursement specialist, a procurement specialist, an environmental specialist, and a social



specialist for the PCU with qualifications, experience, and terms of reference satisfactory to the Association, and (iii) thereafter maintain the PCU, including such specialists throughout Project implementation

Financing Agreement: Schedule 2, Section I.A.1(b). The Recipient, through the SIDWSWD, shall by no later than one hundred eighty days (180) days after the Effective Date, and prior to the commencement of any Project activity at district level, cause the Eligible DWSSP to establish, and thereafter maintain throughout Project implementation, a technical team within each Eligible DWSSP composed of a district project coordinator and water engineer, a sanitation officer, an environmental and social officer, and an institutional development officer, all with qualifications, experience, and terms of reference acceptable to the Association, to be responsible for coordinating with the PCU and overseeing day-to-day Project implementation at the district level.

Financing Agreement: Schedule 2, Section I.A.2. To facilitate the implementation of Part 3.1 of the Project, the Recipient, through the SIDWSWD, shall, by no later than ninety (90) days after the Effective Date, hire, and thereafter maintain throughout the Project implementation, an independent verification agent, under terms of reference and with qualifications and experience acceptable to the Association, to carry out independent verifications, in accordance with the terms of the POM, of each Eligible DWSSP's performance of the service delivery improvement requirements set forth in the respective Performance-based Grant Agreement.

Financing Agreement: Schedule 2, Section I.B.1. For the implementation of Part 3.1 of the Project, the Recipient, through the SIDWSWD, shall ensure that all Performance-based Grants financed out of the proceeds of the Financing are made and administered in accordance with the eligibility criteria and requirements set forth in this Agreement and further detailed in the POM, unless otherwise agreed to by the Association in writing and thereafter set forth in the POM.

Financing Agreement: Schedule 2, Section I.C.1. The Recipient shall, through the SIDWSWD, prepare and adopt the POM in accordance with this Agreement and in a manner satisfactory to the Association, and shall ensure that the Project is carried out in accordance with the POM, setting forth rules, methods, guidelines, and procedures for the carrying out of the Project.

Financing Agreement: Schedule 2, Section I.E.1. The Recipient shall, through the SIDWSWD, ensure that the Project is carried out in accordance with the Environmental and Social Standards, in a manner acceptable to the Association.

Conditions

Type	Citation	Description	Financing Source
Effectiveness	Financing Agreement: Article IV, 4.01(a)	The Recipient, through the SIDWSWD, has adopted the Project Operations Manual ("POM") in form and substance satisfactory to the Association.	IBRD/IDA
Effectiveness	Financing Agreement: Article IV, 4.01(b)	The Co-financing Agreements have been executed and delivered and all conditions precedent to their effectiveness or to the right of the Recipient to make withdrawals under	IBRD/IDA



		them (other than the effectiveness of this Agreement) have been fulfilled.	
Effectiveness	Financing Agreement: Article IV, 4.01(c)	The Grant Agreement has been executed and delivered and all conditions precedent to its effectiveness or to the right of the Recipient to make withdrawals under it (other than the effectiveness of this Agreement) have been fulfilled	IBRD/IDA
Effectiveness	Grant Agreement, Article IV, 4.01(a)	The execution and delivery of this Agreement on behalf of the Recipient have been duly authorized or ratified by all necessary governmental action.	Trust Funds
Effectiveness	Grant Agreement, Article IV, 4.01(b)	The Financing Agreement has been executed and delivered and all conditions precedent to its effectiveness or to the right of the Recipient to make withdrawals under it (other than the effectiveness of this Agreement) have been fulfilled.	Trust Funds
Disbursement	Financing Agreement, Schedule 2, Section III.B.1(b)	No withdrawal shall be made under Category (4), unless and until the Association has received satisfactory evidence that: (i) at least one Performance-based Grant Agreement has been duly executed between the SIDWSWD and the respective Eligible DWSSP under terms and conditions acceptable to the Association; and (ii) an	IBRD/IDA



		independent verification agent has been hired with qualifications and terms of reference satisfactory to the Association and in accordance with paragraph 2 of Section I.A of this Schedule.	
Disbursement	Financing Agreement, Schedule 2, Section III.B.1(c)	No withdrawal shall be made for Emergency Expenditures under Category (6), unless and until all of the following conditions have been met in respect of said expenditures: (i) (A) the Recipient has determined that an Eligible Crisis or Emergency has occurred, and has furnished to the Association a request to withdraw Financing amounts under Category (5); and (B) the Association has agreed with such determination, accepted said request and notified the Recipient thereof; and (ii) the Recipient has adopted the CERC Manual and Emergency Action Plan, in form and substance acceptable to the Association.	IBRD/IDA
Disbursement	Financing Agreement, Schedule 2, Section III.B.1(d)	No withdrawal shall be made for Eligible Expenditures under Category (7), until and unless the Association has notified the Recipient that the conditions set forth in Section 5.15 (a) of the General Conditions have been fulfilled.	IBRD/IDA



Disbursement	Financing Agreement, Schedule 2, Section III.B.1(e)	No withdrawal shall be made for Complementary Financing for the Cat DDO under Category (8), until and unless: (i) the Recipient has furnished to the Association a request to reallocate and thereafter withdraw all or part of the Unwithdrawn Credit Balance for the Complementary Financing for the Cat DDO, and such notice specifies the Cat DDO Legal Agreement; and (ii) the Association has accepted said request and notified the Recipient thereof, and is satisfied, based on evidence satisfactory to it, that the conditions precedent to withdrawal of the financing provided under the Cat DDO Legal Agreement have been fulfilled.	IBRD/IDA
Disbursement	Grant Agreement, Schedule 2, Section II.B.1(b)	No withdrawal shall be made under Category (1), unless and until the Association has received satisfactory evidence that: (i) at least one (1) Performance-based Grant Agreement has been duly executed between the SIDWSWD and the respective Eligible DWSSP under terms and conditions acceptable to the Association; and (ii) an independent verification agent has been hired with qualifications and terms of reference satisfactory to	Trust Funds



		the Association and in accordance with paragraph 2 of Section I.A of this Schedule.	
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I. STRATEGIC CONTEXT

A. Country Context

1. **The Kyrgyz Republic is a lower-middle-income country with strong but volatile economic growth.** Over the last 20 years, the country's economic growth pattern has been uneven, with an average Gross Domestic Product growth rate of 4.5 percent. The economic gains have resulted in a dramatic, though fitful, reduction in poverty, reflected in a decrease in monetary poverty headcount from 36.6 percent in 2013 to 20.1 percent in 2019.¹ However, this trend was reversed by the Coronavirus disease 2019 (COVID-19) outbreak, related economic fall-out, and political upheavals, compounded by high global food and fuel prices, which increased domestic inflation. A third of the 7.1 million inhabitants live below the poverty line.² Although the economic growth reached 9 percent in 2022 (much stronger than expected), driven by construction boom, gold production, agriculture, and the services sector, the country's growth forecast is less optimistic, with a 6.5 percent in 2024 and persistent vulnerability to market fluctuations and external shocks due to the economy's heavy dependence on resource-based commodities and remittances. The country's development objectives outlined in the National Development Strategy 2018-2040³ aspire to a strong, self-sufficient, and developed state by 2040.

2. **The country's economy is highly vulnerable to climate change and heavily dependent on climate-sensitive sectors.** The water, energy, and agricultural sectors are the most vulnerable to climate change and face a wide variety of natural and climate-related hazards, such as droughts, floods, and mudslides. The country ranked 52nd in German Watch's annual Climate Risk Index, and the most at-risk country in the Central Asia region,⁴ and urgently needs a strategic approach toward climate adaptation and resilience. Between 1991 and 2011, the country suffered at least US\$14 million in average annual agricultural losses, almost entirely attributable to drought and water shortages;⁵ thus requiring sustainable water use practices to safeguard livelihoods. The annual average population affected by floods is about 27,000, with an annual average loss of US\$73 million.⁶ Projections indicate that the country could experience up to 5.6°C of warming—above the global average rise of 3.7°C—by 2090s, and the annual probability of severe drought is expected to double from 14 percent in 2020–2039 to 31 percent in 2040–2059.⁷ Climate vulnerability disproportionately affects the poor and vulnerable, with women and children being at the highest risk, especially in rural areas, due to limited access to essential services. Climate change will also increase sanitation-related risks to public health and, with floods and droughts expected to become more intense and frequent, will exacerbate the spread and transmission of water-related diseases through increased exposure to flood waters contaminated with poorly managed sewage.

3. **The infrastructure and services deficit is hindering the Kyrgyz Republic's development.** The country's infrastructure quality is relatively poor compared to its regional peers.⁸ While there has been an increase in investments, leading to universal access to electricity⁹ and significant improvements in water and transport coverage, quality of services remains a major challenge, particularly in rural areas. Further, the infrastructure and service gap between urban and rural areas is often a source of social conflicts, particularly in mining communities,¹⁰ which lack access to basic services. Water and sanitation investments have historically lagged behind, at one percent of the total country investments compared to 54 percent in energy and 39 percent in transport projects (between 2000 and 2018). Although the country's development strategy strongly focuses on infrastructure financing, more investments are needed in climate-vulnerable sectors.

¹ Poverty and Vulnerability in the Kyrgyz Republic. 2022. World Bank

² <https://data.worldbank.org/indicator/SI.POV.NAHC?locations=KG>

³ Kyrgyz Republic. 2018. National Kyrgyz Republic Development Strategy

⁴ https://www.germanwatch.org/sites/germanwatch.org/files/Global%20Climate%20Risk%20Index%202019_2.pdf

⁵ Kyrgyz Republic. 2016. *Third National Communication of the Kyrgyz Republic under the UNFCCC*. Bishkek.

⁶ *Country Risk Profile: Kyrgyz Republic*. 2022. Central Asia Regional Economic Cooperation Program.

⁷ *Climate Risk Profile: Kyrgyz Republic*. 2021. World Bank Group and Asian Development Bank.

⁸ Sustainable Infrastructure for Low-Carbon Development in Central Asia and the Caucasus: Hotspot Analysis and Needs Assessment. 2022. OECD.

⁹ Multiple Indicator Cluster Survey. 2019. National Statistical Committee of Kyrgyz Republic

¹⁰ <https://www.sciencedirect.com/science/article/pii/S2214790X20302847>



4. The country's economic transformation and growth requires a higher investment efficiency and a greater focus on institutions. The country's vision and development objectives are generally inclusive and largely reformist. However, the country's development strategy remains vulnerable to political instability and social unrest. The largely extractive-based economy has not so far led to sustainable and inclusive growth. Future development would require a shift from the current drivers of growth into new, more inclusive, and sustainable sources, such as energy, water, and agricultural resources. However, these sectors suffer from policy uncertainty and institutional capacity gaps that further encroach into the limited fiscal space with subsidized utilities and services. Thus, stronger institutions, a stable regulatory environment, and robust tariff policies are among the key reforms needed to increase investment financing efficiency and harness the transformative potential of infrastructure and services.

B. Sectoral and Institutional Context

5. Water supply and sanitation (WSS) are essential enablers of socio-economic growth and of human capital in the Kyrgyz Republic. Access to WSS is an important contributor to health and education outcomes in the country. Poor access to water supply has been associated with intestinal helminth and protozoa infections,¹¹ and global studies indicate that at least 60 percent of diarrheal deaths are associated with poor access to water, sanitation, and hygiene (WASH).¹² Access to clean water is also a source of social conflict in some Kyrgyz communities, thus significantly affecting local economic activities. Further, access to WSS is among the key factors affecting children's retention in schools, particularly during winter in rural settlements, where extremely low temperatures prevent children from accessing outside toilets and handwashing facilities. The impact of poor WSS is estimated to cost the country US\$120 million annually.¹³

6. The country has made substantial progress in increasing access to WSS in the past two decades. With an estimated 90.8 percent access to basic water services, the absolute number of unserved people decreased by more than two-thirds between 2000 and 2022, due to the persistent water supply infrastructure investments, including in the enabling policies, institutions, and regulations (PIR). Between 2001 and 2013, the Government of the Kyrgyz Republic (GoKR) implemented the Taza-Suu Rural Water Supply Program with support from the World Bank,¹⁴ Asian Development Bank (ADB), and the Department for International Development of the United Kingdom. Though with some implementation challenges due to limited institutional and technical capacity and governance issues, the program invested in close to 550 villages and resulted in significant access to water supply in the rural areas. In the last decade, additional US\$0.5 billion¹⁵ were invested in water supply, resulting in over 287,000 people provided with an improved water source in at least 30 percent of rural settlements in the country. As a result, cumulative 72 percent of the 2,014 villages in the country are now served with functioning water systems, though some expansion is needed. Similar trends have been observed in access to basic sanitation, currently at 92.6 percent.¹⁶ These access improvements have led to a notable progress in public health: according to the Multiple Indicator Survey Cluster Snapshot of 2018, there has been a steady decline in the infant and child mortality rate under five years old (from 28 to 20 deaths per 1,000 children) since 2000, which is highly correlated to WSS access. Thus, further investments and universal access to WSS have the potential to significantly reduce the burden of public health disease on human capital and the country's economy.

7. In parallel, the GoKR carried out a series of capacity-building activities and pilots to enhance the PIR, with extensive technical support from the World Bank. At the policy level, the Water Code was enacted in 2005, the Program for Development of Drinking Water Supply and Sanitation Systems for Populated Localities (PDDWSSSPL) until 2026 was developed, and the design norms and standards ('SNIps') and the WSS Law¹⁷ were updated in 2023 to better align the

¹¹ <https://www.sciencedirect.com/science/article/abs/pii/S0001706X10001816>

¹² <https://www.sciencedirect.com/science/article/pii/S1438463918310484>

¹³ World Bank. 2019. Central Asia: Regional Water Security (internal report). Washington, DC.

¹⁴ US\$37.64 million were provided under the Rural Water Supply and Sanitation Project (RWSSP-1 and 2, P036977; P110267)

¹⁵ Including US\$115.14 million World Bank financing provided under the ongoing SRWSSDP and the CRWSP.

¹⁶ <https://www.sdg6data.org/en/country-or-area/kyrgyzstan>

¹⁷ Pending government approval (Developed under the World Bank-funded SRWSSDP)



sector's institutional framework with the local government structures, promote quality infrastructure development, and enhance WSS service delivery efficiency and cost recovery. Further, a rural WSS information system (SIASAR¹⁸) was introduced, and extensive training programs were implemented through multiple investment projects, targeting key sector institutions at the central and local levels, with more than 1,000 sector professionals trained over the last ten years.

8. However, the lack of reliability and the low quality of services remains challenging, particularly in rural areas. Poor infrastructure performance, low tariffs, and operational inefficiency significantly affect WSS service quality and sustainability. Though addressing the access gap, most of the sector investments are not transformed into sustainable services but enter into a recurrent cycle of reinvestment, requiring significant repairs and replacements, particularly in small towns and rural areas. Of the 839 rural settlements¹⁹ that benefited from investments in the last 20 years, 311 require upgrading, partly due to engineering design failures and construction deficiencies and partly due to poor operational and asset management practices. Data from SIASAR indicate that less than one-third of WSS service providers (municipal or community-led) are fully operational, and only nine percent are financially sustainable.²⁰ Recently established municipal water enterprises (MWEs) perform better than the community drinking water user unions (CDWUUs) but struggle with poor service management due to the lack of qualified staff and high turnover. Further, due to the low tariffs and low willingness to pay for services, the operators' balance sheet in most villages is characterized by high volumes of debt and uncollected arrears from unpaid water bills. This situation is exacerbated by the lack of metering, which leads to poor water consumption practices. As a result, infrastructure performance is relatively low, with high non-revenue water (NRW) (above 50 percent) and related poor water and service quality.

9. The situation with sanitation is dire, and the official figures mask prevailing service access limitations. In rural areas, pit latrines and septic tanks are the most common containment solutions and fall primarily under the responsibility of households, many of whom lack the financial resources²¹ and technical knowledge to construct and safely operate their sanitation facility. Fecal sludge emptying and transport is managed by private or municipal tankers that dispose of the sludge in either an inoperative wastewater treatment facility or a dumping site, with most of the excreta ending up in the environment without treatment. The lack of sanitation systems is among the main contributing factors to poor public health. Some villagers in Osh region reported a high incidence of kidney stones due to the interaction between boreholes and on-site sanitation systems. Schools and health centers are required to have functioning sanitation facilities. However, many of these facilities, usually outdoors, require significant upgrades and lack a sustainable operation and maintenance (O&M) structure. Aiyl Okmotus (AOs)²² are responsible for financing the continued operation (replacement of spare parts, cleaning, and payment of water bills) of these facilities, but they lack the resources to do so.

10. The growing climate vulnerability exacerbates the limited WSS service reliability. Groundwater accounts for more than 85 percent²³ of the water supply in the Kyrgyz Republic, with more than a third of existing reserves currently being exploited. However, with the increasingly intense and prolonged drought combined with depleted glaciers exacerbated by climate change and growing water demand and abstraction, water availability is compromised, particularly during summer, which is becoming hotter and thus more drought prone. In several settlements, water operators have challenges maintaining more than eight hours of supply. Further, the extremely low temperatures during winter, which are becoming frequent and more severe, result in infrastructure failure for water supply and wastewater, leading to prolonged service interruptions. In addition, increasing landslide disaster events cause major infrastructure failure and service disruption. According to the Ministry of Emergency Situations, close to 30 percent of the Kyrgyz Republic settlements are at risk of

¹⁸ Sistema de Información de Agua y Saneamiento Rural, an initiative initiated in the Latin America and Caribbean countries.

¹⁹ Based on SIDDWSWD data.

²⁰ Based on SIASAR data for 2019-2020.

²¹ The implementation of the results-based incentive grant scheme for sanitation improvement in households under the SRWSSDP showed low commitment of the households due to high costs (at least US\$1,000).

²² Local self-governance bodies

²³ Karimov, T. Kh. et al. 2019. Analysis of Groundwater Resources in the Kyrgyz Republic. Journal of Environmental Management and Tourism, [S.I.], v. 10, n. 5, p. 984-990, ISSN 2068-7729.



landslides, with Osh and Jalalabad regions being at the highest risk. In 2024, at least 23 villages in Osh suffered from water supply disruptions due to landslides.²⁴ The lack of sanitation is also a major source of contamination for individual and communal groundwater sources during climate change-exacerbated flood events.

11. The poor performance of WSS services is particularly critical for women, who bear the burden of water provision at the household level, yet they are not an integral part of policy development and decision-making. In the Kyrgyz Republic, as is generally the case in developing countries, women are most likely to be responsible for fetching water for households, and girls are nearly twice as likely as boys to bear the responsibility and spend more time doing it each day (a disparity which could be exacerbated with climate change induced droughts).^{25,26} However, women are underrepresented in the WSS sector in the country. The rapid assessment in 2022 indicated that women comprise 25 percent of staff, 32 percent of staff in technical positions, and only 15 percent in managerial positions in urban utilities. In rural service providers, SIASAR reports that around 30 percent of the staff are women and work primarily as accountants or controllers. There are no women in managerial or technical positions, as confirmed by the evidence from the ongoing World Bank-financed projects. This gender gap is mainly explained by sociocultural norms in what is a male-dominated sector, few female role models in managerial positions, low salaries,²⁷ lower enrollment of women in the science, technology, engineering, and mathematics fields, low access to information and capacity development,²⁸ and the low likelihood of pursuing technical careers.²⁹

12. The country is undergoing a series of PIR reforms to enhance service delivery and accelerate access to WSS. In early 2024, the sector structure was consolidated at the central level to combine policy, strategy development, and investment planning under the new Ministry of Water Resources, Agriculture, and Processing Industry (MWRAPI). A State Institution of Drinking Water Supply and Wastewater Disposal (SIDWSWD)³⁰ under the existing Water Resources Service (WRS) replaced the Department of Drinking Water Supply and Wastewater Disposal (DDWSWD) on the overall sector mandate, including policy development and service regulation. Consultations are ongoing for the proposed delegated service delivery model that aggregates the management, operation, and maintenance of WSS services into district water and sanitation service providers (DWSSPs), with two pilots currently being established in Naryn and Suzak districts through government financing. The sector restructuring is expected to streamline human resources and technical capacity, leverage economies of scale, and promote financial efficiency. The aggregation at the district level will help optimize revenues and finance necessary investments to improve operational efficiency and extend services to unserved areas. These water sector changes are aligned with the ongoing administrative-territorial reform³¹ on the consolidation of AOs, which started in 2022 with pilots in the Issyk-Kul and Naryn regions and halved the number of AOs in 2024.

13. Further PIR improvements are required to reduce the sector's fragmentation, develop the service delivery functions and enhance the climate resilience and financial sustainability of WSS services. Although WSS is now consolidated under a single ministry, the Community Development and Investment Agency (ARIS) still manages a large part of the infrastructure investments, though SIDWSWD is gradually increasing its role in investment management. ARIS also hosts the SIASAR, which supports investment planning and service monitoring. Water tariffs are agreed with the Anti-Monopoly Regulation Service (the entity responsible for economic regulation) under the Ministry of Economy and Commerce and approved by local *Keneshes*³². MWRAPI is in the process of mainstreaming infrastructure investment planning (with the

²⁴ Asian Development Bank. 2023. Atlas of Landslides in the Kyrgyz Republic, 2023.

²⁵ <https://www.who.int/publications/m/item/progress-on-household-drinking-water--sanitation-and-hygiene-2000-2022---special-focus-on-gender>

²⁶ National Statistical Committee and UNICEF. 2019. Kyrgyzstan Multiple Indicator Cluster Survey 2018, Survey Findings Report. Bishkek.

²⁷ Anecdotal evidence suggests that the average salary of the rural service provider staff is 2-3 times less than the average salary of a teacher.

²⁸ Practical Outlook on Gender Issues in the Water Resources Sector (2020). http://cakanonline/wp-content/uploads/collection_eng.pdf

²⁹ World Bank Gender Data Portal. 2018. In the Kyrgyz Republic, only 18 percent of engineering graduates are women, the lowest in Central Asia.

³⁰ The DDWSWD was transferred to the MWRAPI by resolution #98 of the Cabinet of Ministers dated March 7, 2024.

³¹ Decree #370 of the President of the Kyrgyz Republic, dated December 29, 2023.

³² Local councils.



establishment of an investment coordination unit) and playing a role in defining cost recovery tariffs at the DWSSP level. Drinking water quality regulation is under the Ministry of Health. Service provision is decentralized to urban WSS utilities in the cities and district capitals through municipal WSS utilities and in rural areas through MWEs and CDWUUs, many of which lack the capacity to manage WSS services. Thus, it is critical that the regulatory instruments operationalizing the new WSS law, including a clear set of guiding principles for service delegation, cost-reflective tariff setting, and service regulation, are adopted.

14. The GoKR intends to secure potable water supply to all by 2030 and improve sanitation in priority areas in line with its National Development Strategy, the 2018-2026 PDDWSSPL of the Kyrgyz Republic and its successor. At the current investment pace, in the most optimistic scenario, the country would likely cover only 88 percent of rural settlements with basic access to water by 2030. Such a scenario does not consider the risk of falling back, which is high for rural water supply systems with no sound management structures and sustainable operational financing mechanisms. Thus, fully realizing the country's universal WSS access goals requires (i) an accelerated implementation strategy to deliver at least twice the current annual sector investment and (ii) sector reforms focused on service delivery capacity to enhance the sustainability of WSS. Lessons from the implementation of multiple investment programs, including World Bank-financed WSS projects in the country,³³ point to the need for (i) a better alignment between investment models and financing mechanisms to incentivize improvements in operational, financial, and climate sustainability; (ii) a results-oriented institutional capacity development approach to enable skills development on investment planning and implementation, and improve service delivery; and (iii) improvements to the regulatory framework to enable cost-reflective tariffs.

15. Achieving universal access and enhancing climate resilience will require a significant scale-up and consistency in the WSS investments through a program focused on closing the access gap and strengthening the sustainability of prior and new investments within the next ten years. Several development partners, including the Asian Infrastructure Investment Bank (AIIB), the Islamic Development Bank (IsDB), the Organization of Petroleum Exporting Countries (OPEC) Fund, the ADB, and the Swiss Agency for Development and Cooperation (SDC), are willing to step up their engagement and commit to a combined pipeline of US\$200 million and join the World Bank in this transformative program through blended parallel or co-financing arrangements within the fiscal limits for the country. A joint set of program outcomes (basic access to WSS) and an approach to building and sustaining service delivery capacity (results-based) have been agreed upon between the GoKR, the World Bank, and the above-mentioned partners. The World Bank financing will include capacity building and program management interventions to de-risk the Program at all stages, improve readiness, and crowd in more financing as needed. These include preparation and implementation of designs, safeguards, and fiduciary arrangements for the program.

C. Relevance to Higher Level Objectives

16. The proposed investments are aligned with the World Bank's Country Partnership Framework (CPF) for the Kyrgyz Republic for Fiscal Years 2024–2028³⁴. The Project contributes to Objective 2.1 (increase access and efficiency in the water and agriculture sectors). It supports greener and more inclusive growth through investments and reforms targeting WSS services improvement and a reduction in regional disparities. The Project has a poverty focus, including specific design elements to target and extend benefits to the poorest and most vulnerable households, and contributes to reducing gender disparities by improving access and reliability (increasing available hours) of WSS, thus reducing the time women spend collecting water and enabling women and girls to engage in other educational and economic activities.

17. The program is fully aligned with the Paris Agreement and will contribute to the country's development strategy as outlined and supported by the Kyrgyz Republic's national climate policies including the latest Nationally Determined Contribution (NDC). The WSS operational efficiency resulting from this Project's investments will contribute to both mitigation and adaptation actions as outlined in the NDC, which pledges to reduce greenhouse gas (GHG) emissions by

³³ SRWSSDP (P154778); CRWSP (P173734).

³⁴ Report No. 182689-KG, discussed by the Board on October 31, 2023.



15.97 percent below business-as-usual levels by 2030, increasing to 36.61–43.62 percent with international support. In terms of adaptation, the Project will directly contribute to the NDC adaptation measures to improve the climate resilience of WSS infrastructure and stimulate more efficient use of water resources. The Project promotes climate-resilient WSS infrastructure by integrating energy-efficient technologies and solar energy, aligning with the country's Law on Renewable Energy³⁵ to reduce GHG emissions, optimize water management, and strengthen resilience against climate hazards.

18. The program aligns with the Europe and Central Asia Green Transition Framework that calls for a shift towards green, resilient, and inclusive development. The proposed investments are green, as they sustain and restore natural capital by reducing pollution (sanitation and water quality monitoring), improving energy efficiency, and conserving water resources (modernization of infrastructure assets). The proposed investments address resilience by enhancing the reliability of water infrastructure and services and strengthen technical capacity to prepare for the impacts of climatic extremes. The Project also addresses inclusion, as it focuses on enhancing access to WSS for the unserved communities.

19. The program is consistent with the World Bank's vision, fits into its Global Challenge Program (GCP), and advances the achievement of Sustainable Development Goal (SDG) 6 on clean water and sanitation for all. Water security and climate adaptation is the first of the World Bank GCPs that contributes to the creation of a world free of poverty on a livable planet. The water GCP aims to strengthen water security, increase access to safe WSS, and invest in the scale-up of sustainable water management and disaster risk reduction solutions. Within SDG 6, the Project supports the Kyrgyz Republic in advancing towards the following targets: (a) universal and equitable access to safe and affordable drinking water, sanitation, and hygiene (SDG 6.1. and SDG 6.2), and (b) improved water quality by reducing pollution and halving the proportion of untreated wastewater and increasing recycling and safe reuse (SDG 6.3.1). The program also contributes to all three strategic objectives of the WBG new Water Strategy (2025-2030): water for people, water for food, and water for planet.

D. Multiphase Programmatic Approach

(i) Rationale for Using the Multiphase Programmatic Approach (MPA)

20. The universality, sustainability, and climate resilience of WSS services is a long-term pursuit that requires a consistent and focused investment strategy and sector institutional capacity-building. The investment models implemented so far in the Kyrgyz Republic have significantly reduced the WSS infrastructure gap, but the progress on service quality and sustainability remains limited, thus leading to a vicious cycle of infrastructure refinancing. Lessons from the previous and current infrastructure projects demonstrate that the traditional input-based, with a stop-and-redesign approach, limits the effectiveness of institutional capacity-building, which is inherently a dynamic process that requires continuous investment and consistency. A longer-term engagement will incentivize the sector to pursue the critical reforms to enhance the sustainability and climate resilience of WSS and to plan for and finance a more ambitious service-delivery agenda, which is better aligned with the SDGs rather than a narrow focus on infrastructure access.

21. The investment needs to achieve universal access for WSS in the Kyrgyz Republic are significantly higher than the available resources. The GoKR's approach is to leverage World Bank financing to enhance its capacity to attract more resources without worsening its fiscal deficit and, where possible, mobilize private capital and expertise. Thus, the proposed MPA will utilize a programmatic framework with pre-defined infrastructure investment models and service delivery mechanisms³⁶ built from a robust but flexible set of selection criteria that allows for scalability throughout the MPA implementation. Given the institutional capacity gaps and the need to develop a solid financing mechanism to achieve access targets, a long-term partnership is essential to build confidence from other financiers.

22. The proposed Water Supply and Sanitation Universal Access Program (WASUAP) will be operationalized through an MPA using IPF lending instruments with embedded performance-based grants (PBGs). The MPA signals the commitment

³⁵ Kyrgyzstan: [Law on Renewable Energy No. 283 of 2008 \(2019 Edition\)](#)

³⁶ Ready to implement WSS systems, including engineering designs and bidding documents, cost estimate and management model.



of the World Bank and the GoKR to a comprehensive and strategically planned set of investments in the WSS sector, with sustained support contingent on overall implementation performance and achievement of results. The MPA will also incentivize the GoKR to remain focused on effective sector reforms to build the service delivery capacity and customer orientation at the operational level, enable the sustainability of WSS services, and potentially reduce the fiscal burden of poor WSS. It provides flexibility to other development partners and creates an enabling environment for the private sector to invest in different program components. Further, the MPA offers flexibility in adapting the investment program by integrating the learning from the early stages. The combination of IPF and PBGs will enable infrastructure delivery while incentivizing sector reforms, particularly at the district and system levels. The infrastructure packages will prioritize unserved settlements to get to the finish line, and the PBGs will help prevent and mitigate the risk of falling back and ensure long-term sustainability.

(ii) Program Results Chain

23. The program will focus on two key areas of results: (i) increased access to WSS services and (ii) improved sustainability of WSS services, including financial and operational efficiency, climate resilience, and infrastructure financing. The preliminary Program Results Chain is presented in Figure 1. The Theory of Change assumes (i) the willingness of the local government to delegate/enter into a contract with the DWSSP, (ii) the willingness of beneficiaries to pay for cost-recovery tariffs, and (iii) existing national and regional market absorption capacity to double the pace of sector infrastructure development. These elements are largely incorporated into the program design through the PBGs and stakeholder engagement activities.

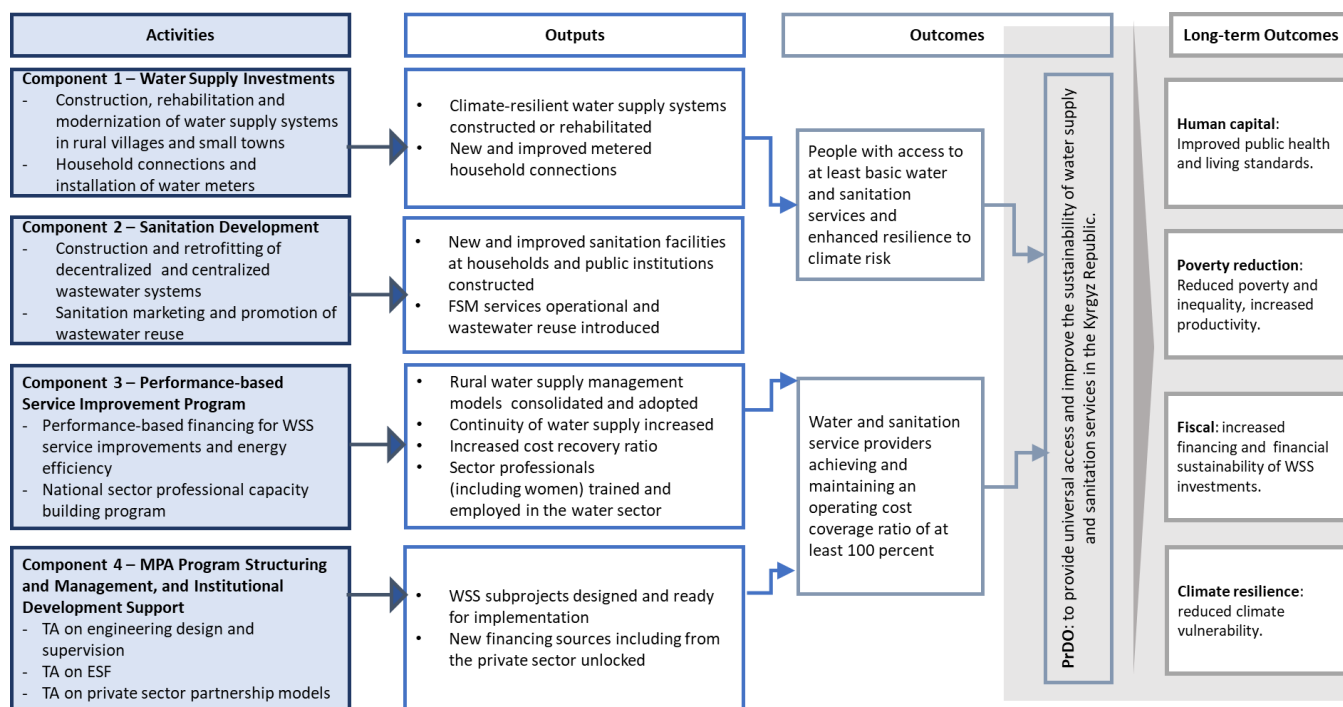


Figure 1. Program Results Chain

(iii) Program Development Objective (PrDO) with Key PrDO Indicators with Baselines and End Targets

24. PrDO statement. The proposed PrDO of the WASUAP is to provide universal access and improve the sustainability of water supply and sanitation services in the Kyrgyz Republic.

25. PrDO Level indicators. At the end of the 10-year Program, the outcomes will be measured by the following indicators:



- **People provided with at least basic water** (of which % is provided with safely managed water) (Baseline: 90.8 percent, target: >99 percent);³⁷
- **People provided with at least basic sanitation** (of which % is provided with safely managed sanitation) (Baseline 92.6 percent, target: **95 percent**);
- **Percentage of the water and sanitation service providers** achieving and maintaining an operating cost coverage ratio of at least 100 percent (Baseline: 9 percent, target: **50 percent**); and
- **People with enhanced resilience to climate risks** in the Kyrgyz Republic as a result of program interventions (Baseline: 0, target: 500,000).

(iv) Program Framework

26. The MPA is structured as a 10-year engagement through three IPF operations/phases to achieve the PrDO. The Project Development Objective (PDO) for each phase and PrDO target the same topics, and the phases are progressive; each phase maintains a similar structure and builds towards the achievement of the overall PrDO. The structure of the MPA is uninterrupted to ensure that the Program's activities continue without a gap in the rollout of the key activities. The total Program's estimated cost is US\$400 million, with a proposed IDA financing envelope of US\$200 million. This includes (i) a **four-year Phase 1 (2025-2029)** of focused investment covering unserved and underserved rural settlements, piloting the aggregated district-level service delivery structure and enhancing the capacity using PBGs, few and simpler sanitation interventions, complemented by extensive analysis and design of more complex sanitation solutions, including paving the way for potential private sector engagement; (ii) a **four-year Phase 2 (2028-2032)** to scale-up the investments, shifting to water supply infrastructure upgrade and expansion, priority sewerage investments and scale-up the service delivery models, with a focus on operational and financial efficiency, and climate adaptation; and (iii) a **four-year Phase 3 (2031-2035)** to scale-up sanitation investments and consolidate the water supply improvements, with an emphasis on knowledge exchange and appropriation of the policies, plans, and tools developed under the previous two phases.

Table 1. MPA engagement

Phase #	Ops ID	Sequential or Simultaneous	Phase's Proposed DO	IPF or PforR	Estimated IBRD Amount (\$ million)	Estimated IDA Amount (\$ million)	Estimated Other Amount (\$ million) ³⁸	Estimated Approval Date	Estimated Environmental and Social (E&S) Risk Rating
1	P500620	n/a	to increase access to WSS services and improve the service delivery capacity in selected areas of the Kyrgyz Republic.	IPF	n/a	121.00	79.00	April 2025	Substantial
2	TBD	Simultaneous	TBD	IPF	n/a	39.00	61.00	June 2028	TBD
3	TBD	Simultaneous	TBD	IPF	n/a	40.00	60.00	June 2031	TBD
Total					n/a	200.00	200.00		
Revised Financing Envelope						\$ 400.00			
Board-Approved Financing Envelope						\$ 200.00			

27. **Phase 1 (US\$200 million, of which US\$121 million IDA financing)** will cover the first batch of the Program's WSS infrastructure investments in unserved and underserved settlements in the least served regions and districts within those

³⁷ The Joint Monitoring Program classifies countries estimated to have achieved >99 percent service coverage as 'nearly universal', recognizing the physical, and economic and financial limitations for last-mile coverage.

³⁸ This includes co-financing as well as parallel financing from development partners.



regions without ongoing or planned investments. Phase 1 will finance the construction and upgrade of water supply systems, and priority household and social institutions' sanitation infrastructure (containment). The WSS interventions will build on experience and lessons learned under the ongoing World-Bank financed Sustainable Rural Water Supply and Sanitation Development Project (SRWSSDP, P154778) and Climate Resilient Water Services Project (CRWSP, P173734) by putting a stronger emphasis on incentivizing service improvements in parallel with infrastructure development to ensure the sustainability of these investments. Thus, Phase 1 will include a PBG-structured capacity-building program for eight DWSSPs to enable them to independently expand and enhance the quality of services within their district. The PBGs will complement ongoing capacity-building programs that target national-level institutions. The Project will also finance improvements to existing fecal sludge management (FSM) services led by both private and municipal operators and the design of sanitation interventions to be implemented in the subsequent phases. Phase 1 envisages the establishment of a learning program to document lessons and inform the program's implementation. Further, the Project will finance program structuring support, including outreach and mobilization, to help attract financing for the subsequent phases.

28. Phase 2 (US\$100 million, of which US\$39 million IDA financing) will scale up the PBG program to other DWSSPs, upgrade and expand existing water supply systems in rural and urban settlements. Under this phase, FSM and household sanitation will be scaled up and complemented by priority sewerage investments using the designs and knowledge developed in Phase 1. Private sector engagement for wastewater improvements will be pursued in partnership with the International Finance Corporation (IFC), targeting settlements around Issyk-Kul Lake, a vital region for the country's tourism industry currently threatened by untreated wastewater discharges. By the end of this phase, it is expected that all the settlements in the country will have access to (at least) basic water services. The PBGs will be expanded, learning from Phase 1, including adjustments to enhance their implementation efficiency. This phase is planned to begin four years after Phase 1 starts, to allow sufficient time to incorporate lessons from Phase 1 and attract additional financiers. However, the program is flexible to an earlier Phase 2 start, subject to the readiness of designs and availability of financing.

29. Phase 3 (US\$100 million, of which US\$40 million IDA financing) will prioritize scaling up sanitation improvements across the country and developing a large-scale investment package to promote safely managed sanitation, targeting the full-service chain and potential waste-to-resource opportunities. Under this Phase, the program will support additional last-mile water supply infrastructure with targeted upgrades and expansion. In addition, this Phase will include consolidating the service delivery model, with a greater focus on investment management capacity for the DWSSPs, innovation, further efficiency improvement, and private sector engagement in water supply. Further, under this Phase, priority will be given to knowledge exchange within regional and DWSSPs to ensure widespread appropriation of the policies, plans, and tools developed under the WASUAP. Phase 3 is expected to start implementation in 2031.

(v) Learning Agenda

30. The Program will use the adaptive learning feature of the MPA to ensure a flow of lessons and knowledge generated through implementation. The nature of the Program presents similar infrastructure and institutional strengthening activities in the three phases, making the MPA a suitable tool for designing innovative approaches, learning lessons, and integrating them in the subsequent phases. Throughout the 10-year MPA Program, the phases will promote knowledge development, exchange between WSS service providers (and between entities involved in WSS more broadly) to disseminate, replicate, and scale up lessons learned. The MPA will also provide an opportunity for cross-learning with other Central Asian countries and elsewhere on district-level aggregated WSS service delivery. Learning will also take advantage of the SDC's technical expertise on WSS, regional and global learning platforms, and events to facilitate knowledge sharing. The design of the phases will be informed by data generated through an enhanced or modified SIASAR information system, studies, and research conducted with lessons learned from previous phases and from complementary projects in the Kyrgyz Republic. The MPA's learning agenda will focus on:

- i. New service provider model – development, testing, and roll-out of DWSSPs.** In Phase 1, the MPA will support the GoKR in establishing DWSSP as a new model for providing WSS services in rural areas, including district centers and



rural settlements. The first phase will pilot the service delivery structure in eight of the ten pilot DWSSP³⁹ while gradually integrating existing MWEs into the DWSSPs. In Phase 2, the DWSSPs will be expanded to cover all 40 districts in the country. Phase 3 will strongly emphasize learning exchange between district DWSSPs in the Kyrgyz Republic and regional peers to ensure widespread appropriation of the knowledge developed through the MPA.

- ii. **Performance improvements – development, testing, and roll-out of performance-based capacity building program.** In Phase 1, the MPA will finance a capacity-building program using PBGs to improve WSS service quality and operational and financial sustainability. This includes preparing and adopting medium-term Performance Improvement Plans, establishing service monitoring systems and customer services with feedback mechanisms in place at the district level, and a functioning financial management system for selected DWSSPs. The program will also facilitate peer to peer learning between Kyrgyz water specialists with successful PBGs. Under Phases 2 and 3, the PBG mechanism will be adjusted and expanded, considering the lessons learned and results from Phase 1.
- iii. **Sanitation Development – developing, piloting, and roll-out of on-site sanitation and wastewater treatment solutions.** Under Phase 1, the MPA will support some research, inclusive stakeholder engagement, and the development of detailed engineering designs of decentralized sanitation systems for both vulnerable households and social institutions while supporting the development and piloting of FSM services and exploring wastewater reuse options. Further, it will support the development and implementation of a sanitation marketing and behavioral change strategy. In Phase 2, the MPA will support investments in innovative sanitation systems and start with the implementation of FSM services based on the knowledge generated in Phase 1. Phase 3 will support scaling up country-wide sanitation improvements, integrating lessons from the previous phases.
- iv. **Professional and vocational excellence in the water sector – testing and roll-out of a Professional and Vocational Development Program (PVDP).** Under Phase 1, the MPA will introduce a certificate-oriented, long-term capacity development program, including a national training center for water supply and sanitation, and scholarships for vulnerable youth, in response to the need to develop and create a pipeline of professional staff and skilled workers (including women in technical and managerial positions) in the sector. This activity is a scale-up from a pilot under SRWSSDP. Phase 1 will primarily target employees of water service providers who benefit from other activities under Phase 1. In Phases 2 and 3, the Program will be fine-tuned and made accessible to all service providers countrywide. These phases will also place a stronger focus on academia to build a pipeline of sector professionals.

II. PROJECT DESCRIPTION

31. **This Phase 1 (hereinafter, Project) will improve resilience to climate change exacerbated droughts and floods, with a focus on three strategic areas:** (i) establish a framework for accelerating WSS infrastructure investments and capacity development in the country; (ii) progressively develop climate-resilient WSS infrastructure and enable water supply access in unserved settlements of the Kyrgyz Republic, and (ii) test and quick start the implementation of a PBG financing mechanism to incentivize service delivery efficiency and enable climate adaptation.

32. **The Project presents an opportunity to advance the implementation of the new institutional framework under the ongoing water sector reform.** A results-based financing approach at the center of the Project capacity-building strategy will ensure a greater focus and prioritization of service quality, customer and citizen engagement, approaches to reduce the gender gap, and the pursuit of climate-smart and decarbonized solutions in WSS. The Project will promote skills development and jobs by attracting, training, and integrating young, skilled people into the WSS service delivery chain.

A. Project Development Objective

33. **PDO Statement.** The project development objective (PDO) is to increase access to water supply and sanitation (WSS)

³⁹ The other two will be financed by the government.



services and improve the service delivery capacity in Selected Areas of the Kyrgyz Republic.

34. PDO Level Indicators. The following PDO indicators will measure the achievement of the first phase objectives and progress toward the PrDO at the end of the Project.

(i) **Outcome 1:** To increase access to WSS services:

- People provided with at least basic water (Number), of which % is (provided with safely managed water, female).
- People provided with at least basic sanitation (Number), of which % is (provided with safely managed sanitation, female).⁴⁰
- People benefiting from climate resilient infrastructure (Number), of which (female, youth).

(ii) **Outcome 2:** To improve the service delivery capacity:

- Water and sanitation service providers with an operating cost coverage ratio of at least 100 percent (Percentage).

B. Project Components

35. The component description presents the scope of the financing confirmed at the time of appraisal and includes a US\$9.0 million grant from SCD, which is being processed in parallel with the Project preparation. Should financing be limited, some of the proposed investments under Phase 1 will be prepared for implementation in the subsequent phases.

36. Component 1 – Water Supply Investments (US\$143.75 million, of which US\$61.55 million IDA credit, US\$8.45 million IDA shorter maturity loan, US\$50.00 million co-financing credit from the AIIB, US\$3.75 million parallel financing grant from AIIB and US\$20.00 million co-financing credit from the OPEC Fund) will finance the construction and upgrade of water supply systems in 126 rural villages and small towns in Chui, Issyk-Kul, and Osh regions, benefitting around **450,000** people. The Project will prioritize the use of gravity systems, energy-efficient pumping equipment, and solar energy for the new water supply schemes (which is around 38 percent of the systems) to minimize GHG emissions. Water source development will target groundwater exploitation to provide a more drought-resilient supply, and network construction will utilize climate-resilient infrastructure design such as materials resistant to high-temperature variations and withstand ground movements in landslide-prone areas in the Osh region. All water intakes will include source protection and reforestation interventions to prevent contamination and enable water conservation and aquifer recharge, further enhancing drought resilience while sequestering carbon. The system design will adopt the World Bank Resilient Water Infrastructure Design Brief (WB-RWIDB)⁴¹ to minimize the risk of water supply disruption due to landslides and other climate-related events. The design will also follow a participatory approach, with consultations involving disabled people, building on local knowledge and historical data. Engineering designs for this component will be funded through the US\$ 3.5 million AIIB grant. Additional water supply system upgrades to ensure universality will be covered through **ADB** financing in **Naryn region (US\$32.00 million serving 36,000 people)**⁴² and **IsDB** financing in **Jalalabad region**.⁴³ Specifically, the Project will finance:

37. Subcomponent 1.1 – Water Supply Access to the Unserved (US\$43.70 million) will finance the design and construction of climate-resilient water supply systems in 48 currently unserved settlements (around **128,000** beneficiaries). Specifically, the Project will finance resilient water source development and storage, thereby reducing drought vulnerability, water treatment infrastructure to ensure high drinking water quality hence boosting positive health outcomes, robust transmission and distribution networks capable of withstanding high temperatures and other climate shocks, reservoirs

⁴⁰ All the sanitation beneficiaries are expected to benefit from water supply improvements. To prevent double counting, these will be counted in the overall number of water beneficiaries. To track progress on sanitation, this indicator is maintained at the PDO level.

⁴¹ Source: <https://documents.worldbank.org/en/publication/documents-reports/documentdetail/868981599035366969/resilient-water-infrastructure-design-brief>

⁴² Approved in September 2024

⁴³ Ongoing feasibility study to determine the number of beneficiaries and financing needs, possibly joining in the Program's Phase 2.



to increase water availability and thus provide drought resilience and metered household connections to ensure sustainable water use.

38. Subcomponent 1.2 – Water Supply Upgrades (US\$100.05 million) will finance the design, rehabilitation, and upgrade of water supply systems in 78 villages and district centers in Aksu, Alai, Aravan, Kara-Kulja, Kara Suu, Panfilov, Sokuluk, and Uzgen districts (around **322,000** beneficiaries). The upgrades will prioritize investments with demonstrated impacts on drought resilience while also improving service reliability and financial viability. These are specific investments focused on source and storage increase to reduce climate-exacerbated water shortages, network hydraulic improvements to reduce water losses, and expansion to secure water supply in the face of climate variability, ensure efficient water demand management, and reduce vulnerability to the impacts of drought.

39. Component 2 – Sanitation Development (US\$29.00 million IDA credit) will finance priority on-site sanitation investments, including household and public sanitation infrastructure, FSM services improvements, and technical assistance (TA) to design more complex sewerage improvements. Specifically, the Project will finance:

(i) The construction, rehabilitation/retrofitting of decentralized sanitation systems, including toilet upgrades for vulnerable households and selected public institutions (schools, kindergartens, hospitals, and health centers), which will ensure accessibility to persons with disabilities in all 126 target settlements, to make them flood resilient. These sanitation improvements are expected to enhance the community's resilience to climate-related health risks and contribute to climate change mitigation by promoting more efficient water use and reducing energy consumption associated with centralized sanitation systems.

(ii) TA (including business development support to small private operators), equipment, and works for FSM services improvement, including fecal sludge collection, transport, treatment, and re-use, which contributes to climate adaptation by enhancing the resilience of sanitation infrastructure to climate impacts and reducing the risk of contamination during floods and to climate mitigation by preventing the release of methane from untreated waste.

(iii) The implementation of a sanitation marketing and behavioral change strategy to incentivize the adoption and use of improved WASH and behaviors that are essential for flood and drought resilience and mitigating the pressures on water resources; and

(iv) TA for the design of sewerage systems in priority settlements, aiming to reduce environmental pollution and enhance the resilience of sewerage infrastructure to the impacts of climate change, such as increased flooding, and extreme weather events. The Project will prioritize the design of sewerage pipelines capable of withstanding harsh weather conditions and wastewater and fecal sludge reuse solutions that enable methane capture for settlements with multi-story buildings, whose sanitation systems are currently a source of soil and groundwater contamination.

40. Component 3 – Performance-based Service Improvement Program (US\$22.00 million, of which US\$13.00 million IDA credit and US\$9.00 million SDC grant) will (i) provide PBGs for DWSSPs to finance WSS service improvements (including NRW reductions) and enhance the sustainability and drought resilience of WSS services, and (ii) finance the professional and vocational development program (PVDP). The PBG allocation per district was estimated using the WSS district aggregation financial model, which considered planned infrastructure improvements, enhanced service levels, and operational efficiency. Additional advisory TA will be provided through a Bank-executed trust fund financed by SDC.

41. Sub-component 3.1 – Service Delivery Improvements (US\$18.00 million) will finance PBGs for eight DWSSPs (out of the ten planned for Phase 1) and will cover the start-up activities and operational and structural investments that are critical to enabling the operational and financial efficiency of the DWSSPs, and the resilience of water supply systems against climate variability. Funding for the PBGs will cover activities from the DWSSPs' institutional and service development plans and will focus on increasing service management capacity, innovative solutions for service expansion and climate resilience, asset management and operations, digital solutions for customer services and feedback, tariff-setting procedures, financial management, and service monitoring and reporting.



42. **The PBG will be structured in two parts:** (i) a fixed part linked to the achievement of institutional prerequisites (the legal establishment of the DWSSPs, adoption of a service improvement plan, a gender empowerment plan, and an approved tariff review framework and an auditable financial management system) to encourage the adoption of sustainable and equitable water management practices that support climate adaptation and climate mitigation by promoting energy-efficient operations and reducing water-related emissions. The fixed grant will only be available during the first 18 months after Project effectiveness; (ii) a variable part linked to service delivery performance, including coverage, hours of supply, customer satisfaction, cost recovery, and NRW reduction, aiming to improve water use efficiency, thus supporting climate mitigation efforts by optimizing resource use and contributing to climate adaptation by ensuring reliable water supply during climate-induced disruptions. The exact set of performance indicators will be defined in the performance agreement between the SIDWSWD and the DWSSPs.

43. **Sub-Component 3.2 – Professional and Vocational Development Program⁴⁴ (US\$4.00 million).** The Project will finance the establishment of a national training center for water supply and sanitation hosted at the SIDWSWD, with a dedicated certificate-oriented, long-term capacity development program in response to the need to create a pipeline of water sector professionals with the skills to plan and manage efficient and sustainable water services. Training packages will be co-developed and implemented with support from the SDC, including adaptive WSS infrastructure and services design and implementation. By equipping professionals in the water sector with the knowledge and skills to implement energy-efficient technologies and practices that reduce GHG emissions, the program will also support climate change mitigation. The PVDP will also carry out extensive outreach and facilitate the enrollment of female participants in training programs designed to attract the youth and women to the water sector and enhance their technical and leadership skills. Training services to support the center will be selected on a competitive basis, considering the technical robustness of the proposed training program, demonstrated training methods, and effectiveness of the monitoring and evaluation (M&E) systems that demonstrate the training impact.

44. **Component 4 – MPA Program Structuring and Management, and Institutional Development Support (US\$9.25 million, of which US\$9.00 million IDA credit, and US\$0.25 million grant parallel financing from AIIB, and in-kind counterpart financing)** will finance TA and institutional capacity building for the establishment of a program management framework and WSS infrastructure development capacity, including:

- (i) TA and training for the development and implementation of the program management tools, including standard E&S instruments, implementation manuals, and protocols, to enhance coordination and program implementation efficiency, and the M&E systems for program results tracking, sector coordination, and annual reviews;
- (ii) TA for the preparation of investment packages, engineering designs, and E&S instruments for future interventions planned for the subsequent phases of the MPA, using WB-RWIDB, including technical assessments, investment screening, and transaction advisory support in collaboration with the IFC to attract private sector engagement in water supply and wastewater solutions along the Issyk-Kul Lake and address vulnerabilities in existing sewerage systems exacerbated by climate change;
- (iii) Incremental operating costs, TA, and equipment for program management;
- (iv) TA, equipment, and services to support a knowledge development program and communications strategy for the MPA aiming to build awareness and understanding among stakeholders about the importance of accelerated WSS access, service delivery sustainability, and climate resilience;
- (v) TA for the preparation of service contract agreements between the operator and asset owner and the PBG independent verification; and

⁴⁴ Introduced through the SRWSSDP-AF1. This activity will be coordinated with the proposed “Water Research Center of Excellence” to be established under the Higher Education Quality and Innovation Project (HEQIP, P178592).



(vi) TA for development and implementation of targeted policy regulations, particularly focused on the implementation of the new service aggregation model and tariff framework.

45. Component 5 – Contingent Emergency Response (CERC – US\$0.00). This component will provide preparedness and rapid response measures to address disaster, emergency, and/or catastrophic events in accordance with the applicable CERC Manual. Following an eligible crisis or emergency event, the Borrower may request the World Bank to reallocate undisbursed Project funds to support emergency response. This component would draw from the undisbursed financing resources under the Project from other Project components to cover eligible crises or emergencies, as needed.

46. Table 2 presents the summary of Project financing (it does not reflect the parallel financing from ADB). In the event that co-financing resources do not materialize, the Project will either (i) be scaled down and the results framework restructured to reflect the revised level of achievements or (ii) pursue additional financing support from IDA if resources become available.

Table 2. Summary of Project Financing (Million US\$)

Component	Co-Financing					Parallel	Total
	IDA	AIIB	OPEC	SDC	Sub-total	AIIB	
Component 1 – Water Supply Investments	70.00	50.00	20.00	-	140.00	3.75	143.75
Subcomponent 1.1: Water Supply Access to the Unserved	21.70	-	20.00	-	41.70	2.00	43.70
Subcomponent 1.2: Water Supply Upgrades	48.30	50.00	-	-	98.30	1.75	100.05
Component 2 – Sanitation Development	29.00	-	-	-	29.00	-	29.00
Component 3 – Performance-based Service Improvement Program	13.00	-	-	9.00	22.00	-	22.00
Subcomponent 3.1: Service Delivery Improvements	9.00	-	-	9.00	9.00	-	18.00
Subcomponent 3.2: Professional and Vocational Development Program	4.00	-	-	-	4.00	-	4.00
Component 4 – MPA Program Structuring and Management, and Institutional Development Support	9.00	-	-	-	9.00	0.25	9.25
Component 5 – Contingent Emergency Response	0.00	-	-	-	0.00	-	0.00
Total	121.00	50.00	20.00	9.00	200.00	4.00	204.00

C. Project Beneficiaries

47. Phase 1 is expected to benefit more than 450,000 people residing in 126 participating rural villages and small towns in Chui, Issyk-Kul, and Osh regions. The majority of beneficiaries within the Project areas will be provided with access to piped water services through a new household connection⁴⁵, and 19,000 people will benefit from household sanitation interventions. Approximately 115,000 people (mostly children) will directly benefit from investments in sanitation facilities and associated hygiene and behavior change interventions in schools, and in other eligible public institutions (for example, health clinics). ADB's parallel project will cover estimated 36,000 people in Naryn region.

48. Beneficiaries of the institutional strengthening activities will include national and local governments and community-level institutions. This includes the SIDWSWD, the Sanitary Epidemiological Surveillance units located within the Ministry of Health at the central and district government levels, the AOs, and the DWSSPs. Ultimately, the customers in the Project areas will benefit from improved service delivery due to the institutional capacity-building activities.

D. Rationale for Bank Involvement and Role of Partners

49. The World Bank is a long-standing partner of the GoKR in supporting its vision to ensure WSS access for all. The Project capitalizes on lessons learned from more than 20 years of WSS engagements in the country and the Bank's invaluable financial, knowledge, and convening services that are critical for the successful implementation of the Project. IDA has also been a catalyst for developing national water sector legislation, including the update of the Water Code of 2005, the current sector strategy, and revisions to the WSS Law. Given the public nature of water, IDA support is one of

⁴⁵ Additional Project beneficiaries will be integrated from Jalalabad during the appraisal of the IsDB funded Project.



the few options available to the Kyrgyz Republic to extend water access to all sustainably and inclusively. Further, the Bank has a track record and a great comparative advantage in the promotion of integrated water resources management.

50. The World Bank will also play a catalytic role in mobilizing additional financing commitments from other development partners and in supporting the technical and safeguards implementation for the program investments. For instance, investments planned under the ADB and IsDB projects will rely on the WBG-designed PBG approach to finance capacity building, and AIIB will rely on the World Bank-funded TA on detailed designs for the water supply network improvements. Further, the SDC expressed interest in supporting TA for institutional capacity building and service development in secondary cities, thus complementing planned investments under the Program.

51. In addition, the Project will finance TA in partnership with IFC to create a private capital-enabling (PCE) environment and help attract competent private partners to finance wastewater treatment in tourist settlements along the Issyk-Kul Lake in line with the WBG's Maximizing Finance for Development (MFD) approach. Given the limited availability of concessional financing compared to the sector investment needs, the program will support TA for preparing and improving the readiness of potential public-private partnerships, targeting settlements with significant economic activity and capacity to pay. Options analysis for potential investments will assess the feasibility of mobilizing capital financing without worsening the constrained fiscal space.

52. Components 2 and 4 are verified as MFD-enabling (MFD-e) and Component 2 is also verified as PCE. Component 2 finances TA and business development support to assist private operators in identifying potential investment opportunities along the FSM chain, as part of the government's plan to increase access to sanitation through additional privately financed FSM services. In partnership with IFC, Component 4 supports the preparation of investment packages, technical assessments and transaction advisory TA which is expected to result in the development of privately financed water supply and wastewater treatment solutions in Issyk-Kul region. Component 2 is also verified as PCM and PCE, given that the business development support, improvements on availability of sludge treatment facilities is expected to enable private investment in FSM.⁴⁶ The PCE following the finalization of Phase 1 is captured in the non-monetary results indicator on the number of operators that invest private capital in FSM services, which is estimated based on the average number of private FSM operators per district, to cover the current service demand.

E. Lessons Learned

53. The WASUAP reflects lessons learned from previous World Bank-financed projects in the Kyrgyz Republic and international experience of WSS programs, including careful consideration of sustainability dimensions. These were reinforced and further elaborated through extensive stakeholder consultations during Project preparation, including a donor roundtable to integrate learning from other interventions and mobilize program support. WASUAP will place greater emphasis on the harmonization of sector investment approaches, aggregation of sector services, and incentives to improve service sustainability. Below is a summary of key lessons reflected in the Project design.

- (i) **The sustainability of WSS services requires aggregation into commercially viable clusters.** Over the last decades, community-based management models have been the preferred strategy of the GoKR for rural WSS. This model relied heavily on community participation and financing and aimed at self-sustainability. In recent years, the limitations of community-based management have become increasingly apparent. The GoKR is now actively looking for alternative models where rural water supply systems are consolidated into larger supply areas and service management responsibilities are extended to several settlements. The aim is to achieve economies of scale, increase the revenue base, reduce overheads, and mitigate the risks associated with infrastructure failure. While achieving universal coverage in rural areas is still a long-term process, international experience shows that aggregation represents an opportunity to tackle the complexity of rural water supply and pave the way for greater efficiency.

⁴⁶ The private investment during Phase 1 is estimated at US\$4 million.



- (ii) **PBGs have proven effective in driving key reforms, including prioritization of resources, financial efficiency, and improved service delivery.** Several World-Bank financed PBGs have demonstrated effectiveness in developing services, including health services in Tajikistan,⁴⁷ local governments in Malawi,⁴⁸ and sanitation services in Mozambique.⁴⁹ In these projects, PBGs were used to incentivize fiscal and service delivery governance and prioritization of resources for critical public services. Results from operations that use PBGs demonstrate that it is an effective financing tool to enhance public services and utility governance systems, improve revenue collection, and enable service delivery, particularly where there are extensive financial constraints. Given the current status of WSS services in the Kyrgyz Republic and the history of input-based financing for capacity building with limited results, the Project will introduce PBGs to incentivize improvements in key financial and operational areas.
- (iii) **Sanitation investments are complex, costly, and require careful planning and assessment.** While the program aims to improve access to both water and sanitation to achieve the full benefits of WSS services, the bulk of the investments in Phase 1 will focus on already designed and well-understood water supply investment models. Sanitation solutions will be studied and gradually implemented throughout the program, starting with simple household and institutional sanitation facilities and FSM services in Phase 1 and moving to complex wastewater conveyance and treatment solutions in Phases 2 and 3. This approach will ensure adequate stakeholder engagement and readiness of infrastructure investments and service arrangements before the implementation.

III. IMPLEMENTATION ARRANGEMENTS

A. Institutional and Implementation Arrangements

54. The institutional and implementation arrangements are built on the experience from previous projects and aligned with ongoing institutional reforms to strengthen the sustainability of service delivery. Key institutions involved in the implementation include the Ministry of Finance (MoF), the MWRAPI, WRS, and its SIDWSWD, DWSSPs, and AOs. The WRS acts as an executive agency, which provides strategic guidance and coordination of the Project, and ensures compliance with domestic procedures as may be required for the Project implementation. The SIDWSWD, as an implementing entity, will lead the Project's implementation, including Program and Project management, and establish the service delivery capacity at the district level, including implementing reforms and the PBGs for the DWSSPs and, where feasible, public-private partnerships. ADB and IsDB projects will rely on ARIS for the implementation of infrastructure components and on the SIDWSWD for capacity building, and PBG for the DWSSPs.

55. The existing Project Implementation Unit (PIU) of the SIDWSWD responsible for the implementation of the ongoing CRWSP will be expanded and further strengthened to manage the proposed program. The PIU will be restructured into a Project Coordination Unit (PCU) at the central level based in Bishkek, with a dedicated team of specialists comprising a Project director for the PCU, and dedicated Project staff, including a manager, a procurement specialist, a disbursement specialist, an environmental specialist, a social development specialist, WSS engineers, an institutional development specialist, M&E specialist, communications specialist, and administrative staff. The PCU will coordinate the management of the PBGs under the SIDWSWD oversight, including the selection of a verification agent. At the district level, technical teams will be established within the DWSSPs to support Project implementation and M&E. The technical teams will include at least a district Project coordinator and water engineer, a sanitation officer, an E&S officer, and an institutional development officer. These technical teams will work closely with the PCU and be responsible for day-to-day Project oversight, including technical support in the implementation of the PBGs and performance improvement plans.

56. Two types of agreements will guide Project implementation. The World Bank, AIIB, and OPEC Fund will sign financing agreements with the MoF as the Recipient of the credits. The World Bank will also sign a grant agreement with the MoF,

⁴⁷ Tajikistan Health Improvement Project (P126130)

⁴⁸ Malawi Governance for Enabling Service Delivery Project (P164961)

⁴⁹ Mozambique Urban Sanitation Project (P161777)



as the grant administrator for SDC. The SIDWSWD will sign performance agreements with each DWSSP, with terms acceptable to the World Bank. All Project operational modalities will be detailed in the Project Operations Manual (POM), which will be adopted prior to Project effectiveness.

B. Results Monitoring and Evaluation Arrangements

57. The Project will rely on the existing SIASAR system, which will be updated, transferred to the SIDWSWD, and expanded to cover district centers. The Project's results framework (RF) and indicators will form the basis for tracking progress on Project outputs and outcomes and will rely on data collected through SIASAR. Process monitoring will rely on existing Project management mechanisms at the PCU and will be enhanced with IT platforms to track critical management processes to achieve the Project's objectives. The PCU will provide biannual progress reports to the World Bank. Procurement and contract management activities will be monitored through the Standard Procurement Documents and Systematic Tracking of Exchanges in Procurement (STEP). The Project will adopt a multi-layered approach to monitoring and results verification at the field level, integrating data generated through the SIASAR, combined with targeted data collection exercises with mobile-based reporting and geo-tagged photos, and periodic field visits. All teams involved in Project implementation will be trained on the use of SIASAR, including data collection, compilation, and analysis.

58. The existing M&E team at the SIDWSWD will be expanded to coordinate all Project M&E activities and strengthened to operate the SIASAR. The split of M&E roles between DWSSPs and PCU, and the reliance on SIASAR will be specified in the POM. Reporting on most Project activities will be undertaken semi-annually to build a learning platform to inform Project implementation. Reports will cover implementation status and results, challenges and proposed actions to address them, procurement and disbursements, and E&S standards. The M&E team will also conduct annual beneficiary surveys for completed sub-projects to assess customer satisfaction. The Project will finance equipment and capacity building, incremental costs to strengthen M&E at the Project level and equip the PCU to carry out these responsibilities. To the extent possible, Project monitoring data will be made publicly available.

C. Sustainability

59. WSS is among the key priorities of the GoKR, and the Project is aligned with the country's WSS strategies. The GoKR is fully committed to the Program and the sustainability of the outcomes. The GoKR has been consistently investing and mobilizing investments in water and sanitation, given the importance of WSS for the well-being of the people and the country's economic growth. The GoKR strategy is to transform WSS service providers into commercially viable entities that can provide high-quality and viable services. The New WSS Law includes provisions to incentivize investments in the sector and expand the services while ensuring a gradual transition into cost recovery.

60. The programmatic approach is structured to pursue service delivery sustainability, and the Project design includes several activities to achieve it. Extensive resources are allocated for institutional capacity building, which is featured as a standalone component of the Project, and the financing for capacity building will be done through PBGs to incentivize a focus on results and enable critical improvements in service governance, operational performance, and financial efficiency. The Project will also finance citizen engagement activities to build support and enhance the accountability mechanisms for WSS services. Business development support will be provided to the local private sector interested in FSM. Further, the Project is deliberately designed to crowd in private-sector financing by designing technical solutions that prepare the ground for private investors. This approach will help enhance commercial discipline in the management of services and encourage the sector to reduce the dependence on donors and government financing.

IV. PROJECT APPRAISAL SUMMARY

A. Technical, Economic, and Financial Analysis

(i) Technical Analysis



61. The Project is designed to build the foundation for a solid program to achieve universal access to water supply and, later, sanitation services. The key Project pillars are: (i) building the program management capacity, investment models and tools to enable the implementation and attract additional financiers; (ii) targeted climate-smart water supply infrastructure to increase access to water supply services focusing on unserved and poorly served settlements; (iii) household and institutional sanitation infrastructure focused on containment while designing robust wastewater solutions for future scale-up; and (iv) results-based financing to incentivize service improvements and enhance the efficiency and sustainability of WSS services. The Project will prioritize a learning agenda focusing on data-driven design, testing, and scale-up of infrastructure and service development solutions. The Project design and the learning strategy rely on strong citizen engagement, which is reflected in the Stakeholder Engagement Plan (SEP). The Project will support the scaling-up of a water sector skills development program that prioritizes integration of female professionals.

62. The Project's rationale, scope, and technical design are based on lessons built over two decades in sector investments and findings from analytical work, including climate risk assessment. The technical design of the infrastructure investments relies on data collected through SIASAR and analytics from engineering designs and works implementation under the previous and ongoing projects. The selection criteria for water supply investments prioritize unserved or underserved settlements in order to ensure basic services while gradually upgrading obsolete infrastructure to ensure high operational efficiency and service quality. Infrastructure interventions will be complemented by capacity-building to enhance sustainability. Detailed cost estimates have been prepared based on unit infrastructure development costs from ongoing World Bank and other projects funded by international financial institutions, complemented by the latest market rates, including the required contingency to cover unforeseen market changes and design complexity.

63. Sanitation investments will start with simpler ones by consolidating the options developed under the ongoing interventions and studying more complex solutions. So far, most sector investments in sanitation have targeted small-scale on-site sanitation focused on household and institutional containment solutions. As access to water supply improves, sewerage services become more critical, particularly in denser settlements such as district centers, where multi-story buildings are commonly used in residential areas. These buildings are serviced by obsolete septic tanks, operating above the designed capacity, with extensive untreated sewage overflowing into the environment and resulting in public health risks. These would require well-structured research and the design of tailored engineering solutions to ensure access to the entire sanitation service chain. Detailed assessments of the latest available technologies will be developed in Phase 1 of the MPA, prioritizing reuse and the technical, financial, environmental, and social feasibility of the proposed solutions.

64. A priority package of water supply investments, comprising 60 percent of the Project's financing, will be designed through the ongoing CRWSP and the AIIB preparation grant. These works, estimated at US\$ 70 million, will be committed in the first 18 months after Project effectiveness as outlined in the 18-month procurement plan (PP). The preparatory activities to enable access to the PBG will also start soon after Project approval to enable its implementation by the time the Project becomes effective. These will focus on the institutional prerequisites to access the fixed start-up grant.

(ii) Economic and Financial Analysis

65. The economic benefits from the Project will derive from improvements in the access and quality of water supply services to households and from sanitation services for households and public institutions. Water supply interventions are expected to result in improved water quality, expanded access, and reduced seasonal variation. These improvements will enhance welfare by reducing coping costs (time saved from water collection and reduced need for in-house water treatment, household storage, and the need to purchase water from tanker trucks) and lead to improved health.

66. An economic analysis was undertaken using estimated Project costs, beneficiary information collected during Project preparation, and benefits observed in similar projects. The analysis incorporates (i) the investment costs of Components 1, 2, and 3 and estimated O&M costs and (ii) all measurable benefits, including decreases in the time spent collecting water, welfare gains at the household level associated with reduced need for in-house treatment (for example, boiling of water), and reduced incidence of water-related diseases such as infectious hepatitis and diarrhea as a result of



improved access to quality water and decline in the reliance on standing water sources. Non-quantified benefits include reductions in household purchases of bottled water, water from tanker trucks, and investments in water storage assets, as well as the improvement in environmental quality resulting from improved sewage management and long-term benefits from improved school attendance by girls resulting from improved sanitation facilities.

67. Based on the above assumptions, a four-and-a-half-year implementation period, and a total projection period of 25 years, the projected net cost-benefit stream yields an Economic Rate of Return (ERR) of 22.7 percent and a Net Present Value, using a five-percent social discount rate of US\$263.9 million. These results likely underestimate the economic returns due to non-incorporation of some current costs of coping and the very low electricity costs in Kyrgyzstan, where the estimates of the economic cost of electricity are far lower than those of comparable countries. Incorporating the reduction in carbon emissions results in an increase in the ERR to 23.0 percent (at the high value of CO2 emissions) and 22.8 percent (at the low value of CO2 emissions). The details of the Economic Analysis are provided in Annex 2.

68. Financial analysis was done for four DWSSPs and will be done for four additional DWSSPs under the Project. An Excel-based model was developed for each DWSSP based on the financial forecast of the consolidated district-level service structure. The output will be a predicted proforma financial statement and a Net Present Value (NPV) until 2035. The model will help determine the financing gap for the service costs and the required PBG to ensure service sustainability.

B. Fiduciary

(i) Financial Management

69. The Project's Financial Management (FM) arrangements have been assessed as *Satisfactory*, and the residual FM risk is *Moderate*. The Project will rely on similar FM arrangements as under the ongoing CRWSP implemented by the SIDWSWD. The existing FM arrangements, including accounting, reporting, planning, budgeting, cash flow and internal controls, and staffing, are assessed to be adequate for the Project implementation. The SIDWSWD continues to meet World Bank requirements under the CRWSP and has significant experience in the implementation of Bank-funded projects. FM arrangements will be reflected in the relevant section of the POM, and the Project will continue to use systems based on existing accounting software, which should also be modified for accounting and financial reporting purposes of the current project. To ensure uninterrupted FM of the Project, an additional Disbursement Specialist will be hired to assist the Finance Manager with Project-related tasks. Quarterly Interim unaudited Financial Reports will be submitted to the Bank by the Recipient through SIDWSWD no later than 45 days after the end of each calendar quarter. Annual audits of the Project's financial statements will be provided to the Bank by the Recipient through SIDWSWD not later than six months after the end of each fiscal year and at Project closure. The Recipient has agreed to disclose the audit reports for the Project within one month of receipt from the auditors and acceptance by the Association by posting them on the SIDWSWD website.

70. Disbursements under the Project will be in line with the World Bank's Disbursement Guidelines for IPF. The Project's designated account (DA) for the credit, in US\$, will be opened and maintained at a financial institution acceptable to the Bank and will be managed by the PCU. The following disbursement methods may be used under the Project: reimbursement, advance, direct payment, and special commitments. The Project will continue to use statements of expenditures as supporting documentation for advances and reimbursements. The ceiling for DA and the detailed instructions on the withdrawal of Project proceeds will be provided in the Disbursement and Financial Information Letter.

71. Funds transferred from SIDWSWD to the DWSSPs for service improvements under Component 3, will be liquidated quarterly. The liquidation report will be supported by appropriate documentation including bank reconciliations documenting actual costs incurred on eligible expenditures under the PBGs. Each DWSSP will open a separate local currency bank account and be subject to internal control policies, procedures, and audits in compliance with local legislation and World Bank-financed operations. Should the DWSSPs spend any funds on ineligible expenditure, they will



not be able to access additional Project funds until all funds for ineligible expenditure are reimbursed and clearance is given by IDA to reinstate the funds flow. A grant manual will be prepared for the PBGs and included in the POM.

(ii) Procurement

72. **The residual Project procurement risk is assessed as *Substantial*.** The World Bank conducted the capacity assessment of the SIDWSWD and found the existing procurement arrangements to be adequate. The PIU under the SIDWSWD has experience in the implementation of projects under the World Bank procedures. However, the assessment identified some risks that could cause procurement delays or inappropriate procurement decisions for the SIDWSWD, as detailed in the risk section, implementation arrangements in Annex 1, and recorded in the Procurement Risk Management Assessment System of the World Bank.

73. **Procurement for the proposed Project shall be carried out in accordance with the World Bank Procurement Regulations for IPF Borrowers, dated September 2023 (Procurement Regulations).** Specific procurement procedures to be followed for managing Project resources will be documented in the POM. The PIU submitted a Project Procurement Strategy for Development (PPSD) and a PP, which have been reviewed and agreed upon with the World Bank before the Project negotiations. The PPCSD is a living document that should be regularly updated during the Project implementation period to provide necessary justifications for procurement arrangements, PP, and its updates. For each contract financed by IDA, the procurement method, market approach, cost estimate, World Bank review requirements, and time frame for implementation shall be agreed between the borrower and IDA and duly reflected in the most updated Project PP. The Project will use the STEP online tool for the planning and tracking of all procurement transactions.

C. Legal Operational Policies

74. **Operational Policy on International Waterways (OP/BP 7.50) will be triggered because the Project will finance the construction, upgrade, and expansion of water supply systems located within the transboundary Chui and Syrdarya river basins.** New construction is expected to require only a low accumulative water abstraction due to the small population size in the unserved settlements. Infrastructure upgrades and water supply service improvements will increase system efficiency, and reduce losses, thereby generating water savings and providing users with a reliable water supply. Water conservation will be promoted through improved demand-management measures, i.e., replacement of continuously running connections, communal standpipes with household connections, and installation of meters. Given the nature of Project activities, the Project will not (i) cause appreciable harm to the other riparian states as it will not adversely change the quality or quantity of water flows, and (ii) will not be appreciably harmed by other riparian states' possible water use. Upon the request from GoKR received on September 11, 2024, the Bank, on behalf of the GoKR, notified the riparian states (Tajikistan, Uzbekistan, and Kazakhstan) on October 10, 2024, and there was no-objection to proceed by riparian countries.

Legal Operational Policies	Triggered?
Projects on International Waterways OP 7.50	Yes
Projects in Disputed Area OP 7.60	No

D. Environmental and Social

75. **The Project's environmental and social risk was assessed as *Substantial* at the Concept stage and remained the same at the Appraisal stage, with *Substantial* for the Environment and *Moderate* for Social.** Environmental and Social



Standards (ESSs) 1, 2, 3, 4, 5, 6, 8, and 10 are considered relevant for the Project. The overall environmental and social impacts of the Project investments are expected to be generally positive. Improved WSS services, including their sustainability and climate change resilience, will benefit the public at large as well as vulnerable groups. They will also have direct benefits to public health through a reduction in the transmission of water-borne diseases. The Project's environmental risks and impacts are mainly associated with the construction of new and rehabilitation of water supply schemes and infrastructure under Component 1 and the construction and/or rehabilitation and/or retrofitting of sanitation systems (i.e., toilet and septic tank upgrades for vulnerable groups and selected public institutions such as schools, kindergartens, health centers, etc.) under Component 2. Such construction is expected to be small and medium scale with site-specific impacts. The Project will utilize groundwater as the main supply, and its use will need to be managed to ensure sustainable withdrawal rates.

76. During the construction phase, potential risks and impacts may include: (i) increased pollution from the generation of hazardous and non-hazardous waste (oil, paints, asbestos-containing materials (ACM) in existing water supply schemes), dust, noise, vibration, and accidental spills of fuel and lubricants; (ii) traffic congestion; (iii) increased suspended sediment and reduced water quality near intakes; (iv) loss of soil material, soil erosion and sedimentation to surface and/or groundwater systems from earthwork activities; (v) improper restoration of construction sites, including the cleaning and sealing of old outdoor unhygienic pits after completion of works; (vi) occupational and community health and safety issues; (vii) disturbance and pollution of natural ecosystems and biodiversity; and (viii) landslides in mountainous terrains. During the operation phase, risks are expected to be mainly related to the wastewater and fecal sludge from sanitation facilities (pathogens, odors, soil, water pollution) and potentially e-waste from discarded solar panels and electric/electronic water meters if not properly managed. Additionally, stress on the aquifer from water intake and supply for the schemes could pose an environmental issue. The aforementioned risks and impacts are typically associated with small- to medium-scale civil works common in water and sanitation interventions. These risks and impacts are largely site-specific and reversible.

77. Project activities related conflicts may emerge in the absence of inclusive and transparent engagement, particularly with communities around the intake areas. There could also be social tension due to the adoption of and/or increased tariffs and required costs to participate (i.e., contributions). No large-scale land acquisition with potentially significant impacts on livelihoods is envisaged under the Project. No significant risks related to labor influx and Sexual Exploitation and Abuse/Sexual Harassment (SEA/SH) are expected under the Project, as most Project workers are expected to be recruited locally. Potential risks during the O&M phase could include community health and safety risks due to increased volume of wastewater and fecal sludge from the decentralized sanitation facilities (pathogens, odors, soil, and water pollution) if not adequately managed; in addition to occupational health and safety risks to the facility operators which are not expected to be complex and significant. The above risks and impacts are readily identifiable and reversible. They can be prevented, minimized, or mitigated by proper assessment and readily available mitigation measures in line with national regulations, the Environmental, Health, and Safety Guidelines of the World Bank, and the Good International Industry Practice. However, considering the type and scope of activities to be implemented in over 100 villages in three regions and the associated ESHS risks and impacts, there could be capacity constraints within the Implementing Agency (IA) in ensuring enforcement and supervision across different contractors. The overall Substantial risk rating also factors in potential capacity constraints and complexity of supervision due to wide geographic coverage.

78. The exact locations and design options will only be determined during the Project implementation following a site-level feasibility assessment. A framework approach to risk management will be applied. This includes the development of an Environmental and Social Management Framework (ESMF), which guides a site-specific assessment of the relevant risks and generic mitigation measures, implementation, and monitoring arrangements that will inform design measures and management of potential impacts. In addition to the ESMF, a Stakeholder Engagement Plan (SEP), Resettlement Policy Framework (RPF), and Labor Management Procedures (LMP) will serve as the main Project instruments for the management of environmental and social risks and impacts across sub-project investments. To that effect, the ESMF, as well as LMP and RPF, were prepared and consulted with relevant stakeholders by the completion of the Project appraisal.



SEP was disclosed prior to the Project appraisal. The ESMF was finalized and disclosed prior to the completion of the Project appraisal. The RPF and LMP were finalized and are expected to be disclosed by March 28, 2025, after translation into the local language. A draft Environmental and Social Commitment Plan (ESCP) was also prepared and was disclosed prior to Appraisal, outlining time-bound actions by the Project IA during Project implementation.

79. The SIDWSWD's E&S capacity will be strengthened by including a central PCU and district-level technical teams, staffed with needed and qualified environmental and social specialists. Capacity requirements for the project will be revisited from time to time during the project implementation.

E. Gender, Citizen Engagement, and Climate Change

80. Gender. The Project narrows two critical gender gaps. First, it seeks to reduce the significant and unequal burden or 'time tax' on women and girls linked to accessing WASH, where there are no options but to seek offsite water sources due to poor supply. Water collection is time-consuming, with 16 percent of women spending between 30 minutes and three hours a day on the task.⁵⁰ The Project will improve reliable access to drinking water supply by increasing the number and hours of service from on-site household water connections under Component 1, freeing up time for productive, educational, and care activities. Annual sample surveys in completed sub-projects will assess the reduction in time spent by women and use for other activities. Second, the Project addresses unequal economic opportunities in WSS sector employment. While 25-30 percent of WSS staff are women, there are no females in technical or managerial positions in the eight DWSSPs. This is partly due to a lack of females with technical training, as the Kyrgyz Republic has only 18 percent female graduates with engineering degrees. Under Component 3, the Project seeks to close the gender gap in the staff composition of WSS utilities by providing women with technical and managerial training under the PVDP and promoting outreach programs and incentives to encourage girls to join available training. The PBGs will also be used to promote gender diversity in the workplace through gender-neutral recruitment policies and procedures, better working conditions, and career opportunities for female graduates. The improvement of the financial sustainability of the utilities is expected to help increase staff salaries, making the sector more attractive for women and the labor force in general. The indicator "Percentage of females who received training in the PVDP and were then hired by the water services sector in technical and managerial roles" is included in the RF.

81. Citizen Engagement. The Project will adopt multiple approaches to enhance transparency, accountability, and social inclusion of the target beneficiaries and interested parties. Learning from previous projects, citizens (including women, youth, and persons with disability) will be engaged in the Project activities through planned consultations and feedback mechanisms at different stages of the Project's cycle. The Project will engage communication tools such as customer service centers, media platforms, and community outreach meetings to respond to inquiries raised by concerned stakeholders. More specifically, under Components 1 and 2, local community members have been consulted through the ESF development process and will be further engaged during implementation through DED, monitoring, and evaluation/lesson learning. Under Component 2, the sanitation marketing and behavior change will rely on the extensive engagement of different beneficiary groups through the human-centered design, messaging design, and dissemination, including peer support in adopting improved sanitation practices and hygiene behavior. As part of Component 3, the Project promotes customer engagement and service-oriented management that includes customer feedback.

82. A SEP has been prepared to guide citizen engagement and defines appropriate mechanisms to engage with both direct and indirect Project beneficiaries and other concerned parties, including vulnerable groups. A beneficiary feedback indicator focusing on service quality, customer orientation and feedback was added to the RF - customer satisfaction index – and will be monitored annually to ensure a greater focus on citizen engagement. These

⁵⁰ National Statistical Committee and UNICEF. 2019. Kyrgyzstan Multiple Indicator Cluster Survey 2018, Survey Findings Report. Bishkek. The report shows that women bear a greater burden for collection of water across all income groups, with the largest gap among the lowest wealth quintile (in 64 percent of the poorest households, women are primarily responsible, versus only 15 percent of households with men being the primary collector), with 12 percent having the female child responsible versus 6 percent where it is the male child responsible."



information/awareness-building processes will be supplemented by a grievance redress mechanism (GRM) that covers all aspects of Project implementation, including grievances related to involuntary resettlement. The GRM data will be collected, compiled, and reported in quarterly reports and discussed during Project implementation support missions, based on which any project procedures causing potential harm to beneficiaries will be adjusted.

83. The Project is aligned with the goals of the Paris Agreement on both mitigation and adaptation. The Project addresses *mitigation risks* across various components by focusing on reducing GHG emissions and enhancing resource efficiency. All Project's activities fall under the universally aligned list. For water supply investments (Component 1), the Project incorporates energy-efficient technologies and solar energy solutions, alongside gravity systems that minimize energy use, particularly in pumping operations. In sanitation development (Component 2), the Project mitigates GHG emissions by preventing methane release from untreated waste through decentralized sanitation systems and fecal sludge management services. The reuse of treated waste as fertilizer decreases the reliance on synthetic fertilizers, further contributing to GHG emission reductions. The PBGs (Component 3) promote efficient water management and service delivery, aiming to reduce NRW and improve overall resource use efficiency. Collectively, these measures not only align with climate mitigation goals but also enhance the Project's capacity to reduce carbon emissions effectively. The Project also addresses the Project area's high vulnerability to droughts and floods (*adaptation risks*) by enhancing resilience against climate change impacts and climate variability. Water supply infrastructures will be designed⁵¹ with materials resistant to high-temperature variations and capable of withstanding floods, landslides, and other climate shocks, ensuring continuous service despite climate-induced variability in water availability. The service improvement program enhances the capacity of water sector institutions to adaptively manage water resources in the face of increased droughts, ensuring sustainable service delivery and water scarcity resilience. Improved sanitation will also increase resilience against exposure to waterborne illnesses via contaminated flood waters. The management component includes training and TA for climate resilience planning, ensuring that long-term sustainability of WSS services as climatic conditions change. Eligible activities included in the CERC and the POM will be Paris-aligned. In all, these climate change risk mitigation measures will reduce residual risks to a low level and are not likely to have a material impact on the operation and its development objectives.

V. GRIEVANCE REDRESS SERVICES

84. *Grievance Redress.* Communities and individuals who believe that they are adversely affected by a project supported by the World Bank may submit complaints to existing project-level grievance mechanisms or the Bank's Grievance Redress Service (GRS). The GRS ensures that complaints received are promptly reviewed in order to address project-related concerns. Project-affected communities and individuals may submit their complaint to the Bank's independent Accountability Mechanism. The Accountability Mechanism houses the Inspection Panel, which determines whether harm occurred or could occur as a result of the Bank's non-compliance with its policies and procedures, and the Dispute Resolution Service, which provides communities and borrowers with the opportunity to address complaints through dispute resolution. Complaints may be submitted to the Accountability Mechanism at any time after concerns have been brought directly to the attention of Bank Management and after Management has been given an opportunity to respond. For information on how to submit complaints to the Bank's GRS, visit <http://www.worldbank.org/GRS>. For information on how to submit complaints to the Bank's Accountability Mechanism, visit <https://accountability.worldbank.org>.

VI. KEY RISKS

85. The overall residual risk is rated Substantial, deriving from substantial political and governance, sector strategies and policies, institutional capacity for implementation and sustainability risks, fiduciary, and E&S residual risks. The Project design includes risk mitigation measures building upon the experience from the ongoing/ previous projects.

⁵¹ These designs are in accordance with the (soft and hard) measures recommended by the [Resilient Infrastructure Design Brief](#)



86. The Political and Governance risk is rated Substantial. Key risks include: (i) the potential impact of recurrent changes in the government systems on sector decision-making and project operations and the lengthy project ratification process by the parliament, which may affect Project effectiveness and implementation; and (ii) risk associated with the operating environment and, in particular, the risk of fraud in the award and implementation of works contracts, which are common to World Bank-financed operations in the infrastructure sector and are considered as general portfolio issues. To mitigate these risks, (i) the preparation phase included several engagements with the relevant government stakeholders at different levels and early agreement on the Project design. This practice will continue as part of the parliamentary ratification process with the members of the parliament once the Project is approved by the Board. (ii) Increased verification and oversight of bidding documents and processes to identify unqualified bidders prior to contract award and increased technical and fiduciary supervision to identify areas of concern early.

87. The Sector Strategies and Policies risk rating is Substantial. Although these are positive changes, the implementation of the ongoing sector reforms, such as the formation of the new MWRAP, may pose a challenge to Project implementation, as it requires multiple stakeholder coordination and may lead to delays in approval procedures for changes in the regulatory framework, and enforcement of water sector strategies, policies, and regulations. These risks will be mitigated with extensive technical support on institutional capacity building and incentives to maintain the focus on institutional reforms through the PBGs, complemented by systematic and sustained stakeholder engagement.

88. The Institutional capacity for implementation and sustainability risk is rated Substantial. The Project will involve a significant increase in sector investment activities and potentially put pressure on the local construction market. Furthermore, the PBG introduced under the Project for the newly established district DWSSPs is an untested innovation in the water sector in the country. The implementation arrangements will be strengthened with a reinforced PCU and technical teams to facilitate implementation on the ground, close supervision of service providers, and engagement with the local government authorities. TA and independent verification will be mobilized to support the DWSSPs and the PBGs implementation to enhance the sustainability of the investments. The engineering design of the first year of investments will be prepared during Project preparation under the ongoing CRWSP. Additional technical and advisory support will be provided by the WBG team on key sector reforms to enhance the institutional capacity and sustainability of the sector.

89. The Fiduciary risk is rated Substantial. Prevalent issues relate to weak procurement and contract management capacity, including (i) price escalation due to unstable market conditions, (ii) attempts by political figures to influence procurement decisions, (iii) limited design and procurement capacity; (iv) local currency depreciation impacts on willingness to submit bids in the local currency; (v) overall unstable procurement environment and frequent changes of procurement legislation, and high level of corruption as measured by Transparency International⁵². Risk mitigation measures include extensive training to strengthen the procurement and contract management capacity; a clear definition of decision-making processes, accountability, and integrity standards for procurement decisions (included in the POM); regional market outreach, strong internal control system for FM and disbursement; and close supervision by the Bank.

90. The Environmental risk is Substantial, and the Social risk is Moderate for an overall E&S risk rating of Substantial. The main E&S risks and mitigation measures will revolve around the activities of Components 1 and 2. The Project's environmental risk rating is based on risks potentially arising from infrastructure modernization and possible water and soil pollution risks. The risks will be mitigated through the preparation, implementation, and monitoring of E&S instruments and training of Project staff and service providers.

⁵² Source: <https://www.transparency.org/en/countries/kyrgyzstan>



VIII. RESULTS FRAMEWORK AND MONITORING

PDO Indicators by PDO Outcomes

Baseline	Period 1	Period 2	Period 3	Period 4	Closing Period
Increase access to WSS services					
People provided with at least basic water (Number)					
Dec/2024	Dec/2025	Dec/2026	Dec/2027	Dec/2028	Dec/2029
0	0	15,000	90,000	247,500	450,000
➤ of which % is provided with safely managed water (Percentage)					
0	0	99	99	99	99
➤ of which % is female beneficiaries (Percentage)					
Dec/2024	Dec/2025	Dec/2026	Dec/2027	Dec/2028	Dec/2029
0	0	50	50	50	50
People benefiting from climate resilient infrastructure (Number of people) ^{CRI}					
Dec/2024	Dec/2025	Dec/2026	Dec/2027	Dec/2028	Dec/2029
0	0	15,000	56,000	140,250	255,000
➤ People benefiting from climate resilient infrastructure - Female (Number of people) ^{CRI}					
Dec/2024	Dec/2025	Dec/2026	Dec/2027	Dec/2028	Dec/2029
0	0	7,000	28,000	70,125	127,500
➤ People benefiting from climate resilient infrastructure - Youth (Number of people) ^{CRI}					
Dec/2024	Dec/2025	Dec/2026	Dec/2027	Dec/2028	Dec/2029
0	0	2,500	16,800	42,000	76,500
People provided with at least basic sanitation (Number)					
Dec/2024	Dec/2025	Dec/2026	Dec/2027	Dec/2028	Dec/2029
0	0	1,000	3,800	10,450	19,000
➤ of which % is provided with safely managed sanitation (Percentage)					
Dec/2024	Dec/2025	Dec/2026	Dec/2027	Dec/2028	Dec/2029
0	0	99	99	99	99
➤ of which % is female beneficiaries (Percentage)					
Dec/2024	Dec/2025	Dec/2026	Dec/2027	Dec/2028	Dec/2029



0	0	50	50	50	50
Improve the service delivery capacity					
Water and sanitation service providers with an operating cost coverage ratio of at least 100 percent (Percentage)					
Dec/2024	Dec/2025	Dec/2026	Dec/2027	Dec/2028	Dec/2029
0	0	3	20	50	80

Intermediate Indicators by Components

Baseline	Period 1	Period 2	Period 3	Period 4	Closing Period
Water Supply Investments					
Villages with water supply systems constructed or rehabilitated with climate-resilient design standards (Number)					
Dec/2024	Dec/2025	Dec/2026	Dec/2027	Dec/2028	Dec/2029
0	0	3	27	69	126
Continuity of water supply in participating water service providers (Hours)					
Dec/2024	Dec/2025	Dec/2026	Dec/2027	Dec/2028	Dec/2029
8	8	8	12	16	20
New metered piped household water connections (Number)					
Dec/2024	Dec/2025	Dec/2025	Dec/2026	Dec/2027	Dec/2029
0	0	6,000	18,000	49,500	90,071
>in Project areas affected by droughts or mudflows (Percentage)					
Dec/2024	Dec/2025	Dec/2026	Dec/2027	Dec/2028	Dec/2029
0	0	1	10	25	56
Sanitation Development					
Households with new or improved sanitation facilities (Number)					
Dec/2024	Dec/2025	Dec/2026	Dec/2027	Dec/2028	Dec/2029
0	0	95	660	1,815	3,300
Social institutions with new or improved sanitation facilities (Number)					
Dec/2024	Dec/2025	Dec/2026	Dec/2027	Dec/2028	Dec/2029
0	0	5	78	214	389
Operators that invest private capital in FSM services (Number)					
Dec/2024	Nov/2025	Nov/2026	Nov/2027	Nov/2028	Dec/2029
0	0	0	5	10	16
Performance-based Service Improvement Program					
Legally created and fully operational district water and sanitation service providers (Number)					



Dec/2024	Dec/2025	Dec/2026	Dec/2027	Dec/2028	Dec/2029
0	0	1	3	6	8
Water supply systems that maintain their non-revenue water below 30 percent (Percentage)					
Dec/2024	Dec/2025	Dec/2026	Dec/2027	Dec/2028	Dec/2029
0	0	1	20	50	100
People who received training in the Professional and Vocational Development Program who are then hired in technical or managerial roles in the water services sector (Number)					
Dec/2024	Nov/2025	Nov/2026	Nov/2027	Nov/2028	Dec/2029
0	0	20	40	60	80
➤Of which female (Percentage)					
Dec/2024	Nov/2025	Nov/2026	Nov/2027	Nov/2028	Dec/2029
0	0	10	20	30	30
MPA Program Structuring and Management, and Institutional Development Support					
Villages with water supply subprojects designed and ready for implementation (Number)					
Dec/2024	Dec/2025	Dec/2026	Dec/2027	Dec/2028	Dec/2029
0	0	0	0	15	30
Communication and citizen engagement strategy developed and adopted (Yes/No)					
Dec/2024	Dec/2025	Dec/2026	Dec/2027	Dec/2028	Dec/2029
No	No	Yes	Yes	Yes	Yes
Customer satisfaction index (Percentage)					
Dec/2024	Dec/2025	Dec/2026	Dec/2027	Dec/2028	Dec/2029
0	0	50	60	70	80
Contingent Emergency Response					

Monitoring & Evaluation Plan: PDO Indicators by PDO Outcomes

Increase access to improved WSS services	
People provided with at least basic water (Number)	
Description	At least basic drinking water means it is from an improved source, provided collection time is not more than 30 minutes for a roundtrip including queuing. Improved water sources include piped household connections (house or yard connections) and community water points (public standpipes, boreholes, protected dug wells, and protected spring and rainwater collection). It does not include water provided through tanker trucks or vendors; unprotected wells or unprotected springs; surface waters (such as rivers, ponds, dams, lakes, streams, and irrigation channels); or bottled water. The definition of "improved water sources" follows the UNICEF-WHO Joint Monitoring Program definition. (WBG Scorecard Corporate Results Indicators)
Frequency	Biannual
Data source	PCU Project Progress reports
Methodology for Data	The indicator will be calculated by adding the number of people at households in the Project areas provided with access to at least basic drinking water as



Collection	a result of Project interventions.
Responsibility for Data Collection	PCU
➤ of which % is provided with safely managed water (Percentage)	
Description	Safely managed means water from an improved water source that is accessible on premises, available when needed and free from fecal and priority chemical contamination. For global monitoring purposes the microbiological standard applied is that no Escherichia coli should be detected in a 100 mL sample and priority chemical contaminants are arsenic and fluoride. <i>(WBG Scorecard Corporate Results Indicators)</i>
Frequency	Biannual
Data source	PCU Project Progress reports
Methodology for Data Collection	The indicator will be calculated by calculating the percentage of people at households in the Project areas provided with access to safely managed drinking water as a result of Project interventions.
Responsibility for Data Collection	PCU
➤ of which % is female beneficiaries (Percentage)	
Description	This indicator measures the percentage of female beneficiaries out of the total number of people provided with at least basic water as a result of Project interventions. <i>(WBG Scorecard Corporate Results Indicators)</i>
Frequency	Biannual
Data source	PCU Project Progress reports
Methodology for Data Collection	The indicator will be calculated by calculating the percentage of female in the Project areas provided with at least basic water as a result of Project interventions, obtained from PCU surveys.
Responsibility for Data Collection	PCU
People benefiting from climate-resilient infrastructure (Number of people) ^{CRI}	
Description	This indicator is defined as the number of people directly benefiting from improved climate risk management and increased climate resilience due to eligible investments during the intervention period. <i>(WBG Scorecard Corporate Results Indicators)</i>
Frequency	Biannual
Data source	PCU Project Progress reports
Methodology for Data Collection	Data will be compiled by the PCU.
Responsibility for Data Collection	PCU
➤ People benefiting from climate resilient infrastructure - Female (Number of people) ^{CRI}	
Description	This indicator is defined as the percentage of female (out of the total number of people) directly benefiting from improved climate risk management and increased climate resilience due to eligible investments during the intervention period. <i>(WBG Scorecard Corporate Results Indicators)</i>
Frequency	Biannual
Data source	PCU Project Progress reports
Methodology for Data Collection	Data will be compiled by the PCU.



Collection	
Responsibility for Data Collection	PCU
➤ People benefiting from climate resilient infrastructure - Youth (Number of people) ^{CR1}	
Description	This indicator is defined as the percentage of youth (out of the total number of people), based on the UN definition of youth (ages 15-24), directly benefiting from improved climate risk management and increased climate resilience due to eligible investments during the intervention period. (WBG Scorecard Corporate Results Indicators)
Frequency	Biannual
Data source	PCU Project Progress reports
Methodology for Data Collection	Data will be compiled by the PCU.
Responsibility for Data Collection	PCU
People provided with at least basic sanitation (Number)	
Description	At least basic sanitation means the use of improved facilities which are not shared with other households. Improved sanitation facilities include flush or pour-flush to a piped sewer system, septic tank, or pit latrine; ventilated improved pit latrine; pit latrine with slab; and composting toilet. It does not include flush or pour-flush to elsewhere (that is, not to a piped sewer system, septic tank, or pit latrine); pit latrine without slab/open pit; bucket; hanging toilet or hanging latrine; shared facilities of any type; or no facilities, bush, or field. The definition of "improved sanitation facilities" follows that of the UNICEF-WHO Joint Monitoring Program. (WBG Scorecard Corporate Results Indicators)
Frequency	Biannual
Data source	PCU Project Progress reports
Methodology for Data Collection	The indicator will be calculated by adding the number of people at households in the Project areas provided with access to at least basic sanitation facilities as a result of Project interventions, obtained from PCU surveys.
Responsibility for Data Collection	PCU
➤ of which % is provided with safely managed sanitation (Percentage)	
Description	Safely managed sanitation means the use of improved facilities that are not shared with other households and where excreta are safely disposed of in situ or removed and treated offsite. (WBG Scorecard Corporate Results Indicators)
Frequency	Biannual
Data source	PCU Project Progress reports
Methodology for Data Collection	The indicator will be calculated by calculating the percentage of people at households in the Project areas provided with access to safely managed sanitation as a result of Project interventions, obtained from PCU surveys.
Responsibility for Data Collection	PCU
➤ of which % is female beneficiaries (Percentage)	
Description	This indicator measures the percentage of female beneficiaries out of the total number of people provided with at least basic sanitation as a result of Project interventions. (WBG Scorecard Corporate Results Indicators)
Frequency	Biannual



Data source	PCU Project Progress reports
Methodology for Data Collection	The indicator will be calculated by calculating the percentage of female in the Project areas provided with at least basic sanitation as a result of Project interventions, obtained from PCU surveys.
Responsibility for Data Collection	PCU
Improve the service delivery capacity	
Water and sanitation service providers with an operating cost coverage ratio of at least 100% (Percentage)	
Description	This indicator measures the percentage of water and sanitation service providers in the Project areas benefitting from Project interventions with an operating cost coverage ratio of at least 100 percent.
Frequency	Biannual
Data source	PCU Project Progress reports
Methodology for Data Collection	OCC ratio = Operational revenues / operational cost. The data will be obtained from PCU surveys.
Responsibility for Data Collection	PCU

Monitoring & Evaluation Plan: Intermediate Results Indicators by Components

Component 1: Water Supply Investments	
Villages with water supply systems constructed or rehabilitated with climate-resilient design standards (Number)	
Description	This indicator measures the total number of villages with water supply systems which have been constructed or rehabilitated with climate-resilient design standards under the Project.
Frequency	Biannual
Data source	PCU progress reports
Methodology for Data Collection	PCU progress reports on completed detailed designs and works.
Responsibility for Data Collection	PCU
Continuity of water supply in participating water service providers (Hours)	
Description	This indicator measures the continuity of water supply services delivered in the Project areas expressed in hours per day.
Frequency	Annual
Data source	PCU progress reports and reports from service providers
Methodology for Data Collection	Data will be compiled by the PCU based on surveys and reports from the participating water service providers.
Responsibility for Data Collection	PCU
New metered piped household water connections (Number)	
Description	This indicator measures the cumulative number of new piped metered household water connections installed under the Project.



Frequency	Biannual
Data source	PCU
Methodology for Data Collection	Data will be compiled by the PCU with inputs from service providers.
Responsibility for Data Collection	PCU
➤in Project areas affected by droughts or mudflows (Percentage)	
Description	This sub-indicator measures the percentage of new metered household water connections installed in the Project areas which are affected by at least one drought or mudflow event during Project implementation.
Frequency	Biannual
Data source	PCU progress reports
Methodology for Data Collection	Data will be compiled by the PCU with inputs from service providers.
Responsibility for Data Collection	PCU
Component 2: Sanitation Development	
Households with new or improved sanitation facilities (Number)	
Description	This indicator measures the cumulative number of households in the Project areas with new or improved household sanitation facilities as a result of Project interventions.
Frequency	Biannual
Data source	PCU progress reports
Methodology for Data Collection	Data will be compiled by the PCU based on the progress of advancement of civil works.
Responsibility for Data Collection	PCU
Social institutions with new or improved sanitation facilities (Number)	
Description	This indicator measures the cumulative number of social institutions in the Project areas with new or rehabilitated sanitation facilities as a result of Project interventions. Social institutions include, but are not limited to, schools, kindergartens, health centers, and hospitals.
Frequency	Biannual
Data source	PCU progress reports
Methodology for Data Collection	Data will be compiled by the PCU based on the progress of advancement of civil works.
Responsibility for Data Collection	PCU
Operators that invest private capital in FSM services (Number)	
Description	This indicator measures the cumulative number of operators in the Project areas that invest private capital in FSM services.
Frequency	Biannual



Data source	PCU progress reports
Methodology for Data Collection	Data will be compiled by the PCU based on the inputs from the DWSSPs.
Responsibility for Data Collection	PCU
Component 3: Performance-based Service Improvement Program	
Legally created and fully operational district water and sanitation service providers (Number)	
Description	This indicator measures the number of district water and sanitation service providers which are established and become fully operational as a result of Project interventions under the Project.
Frequency	Biannual
Data source	PCU progress reports
Methodology for Data Collection	Data will be compiled by the PCU with inputs from service providers.
Responsibility for Data Collection	PCU
Water supply systems that maintain their NRW below 30 percent (Percentage)	
Description	This indicator measures the percentage of water supply systems constructed or rehabilitated under the Project that maintain their NRW below 30 percent.
Frequency	Biannual
Data source	PCU Project Progress reports
Methodology for Data Collection	NRW = Water supplied - water sold (expressed as a percentage of net water supplied). The data will be obtained from PCU surveys.
Responsibility for Data Collection	PCU
People who received training in the Professional and Vocational Development Program who are then hired in technical or managerial roles in the water services sector (Number)	
Description	This indicator measures the number of people who received training in the Professional and Vocational Development Program and who were then hired in technical or managerial roles in the water services sector (e.g., utility, government agency, private sector, etc.).
Frequency	Annual
Data source	PCU Project reports
Methodology for Data Collection	Data will be compiled by the PCU with inputs from service providers.
Responsibility for Data Collection	PCU
>Of which female (Percentage)	
Description	This indicator measures the percentage of women who received training in the Professional and Vocational Development Program and who were then hired in technical and managerial roles in the water services sector (e.g., utility, government agency, private sector, etc.)
Frequency	Annual



Data source	PCU Project reports
Methodology for Data Collection	Data will be compiled by the PCU with inputs from service providers.
Responsibility for Data Collection	PCU
Component 4. MPA Program Structuring and Management, and Institutional Development Support	
Villages with water supply subprojects designed and ready for implementation (Number)	
Description	This indicator measures the cumulative number of WSS sub-projects with completed detailed engineering designs and which are ready for works implementation under Phase 2.
Frequency	Biannual
Data source	PCU Project reports
Methodology for Data Collection	Data will be compiled by the PCU with inputs from service providers.
Responsibility for Data Collection	PCU
Communication and citizen engagement strategy developed and adopted (Yes/No)	
Description	This indicator measures if a communication and engagement strategy has been developed and adopted.
Frequency	Biannual
Data source	PCU Project reports
Methodology for Data Collection	Data will be compiled by the PCU with inputs from service providers.
Responsibility for Data Collection	PCU
Customer satisfaction index (Percentage)	
Description	This indicator measures the percentage of surveyed customers who are satisfied with the quality of water supply services provided and the citizen engagement process.
Frequency	Biannual
Data source	PCU Project reports
Methodology for Data Collection	Data will be compiled by the PCU based on customer satisfaction surveys.
Responsibility for Data Collection	PCU
Component 5. Contingent Emergency Response	



ANNEX 1: Implementation Arrangements and Support Plan

COUNTRY: Kyrgyz Republic

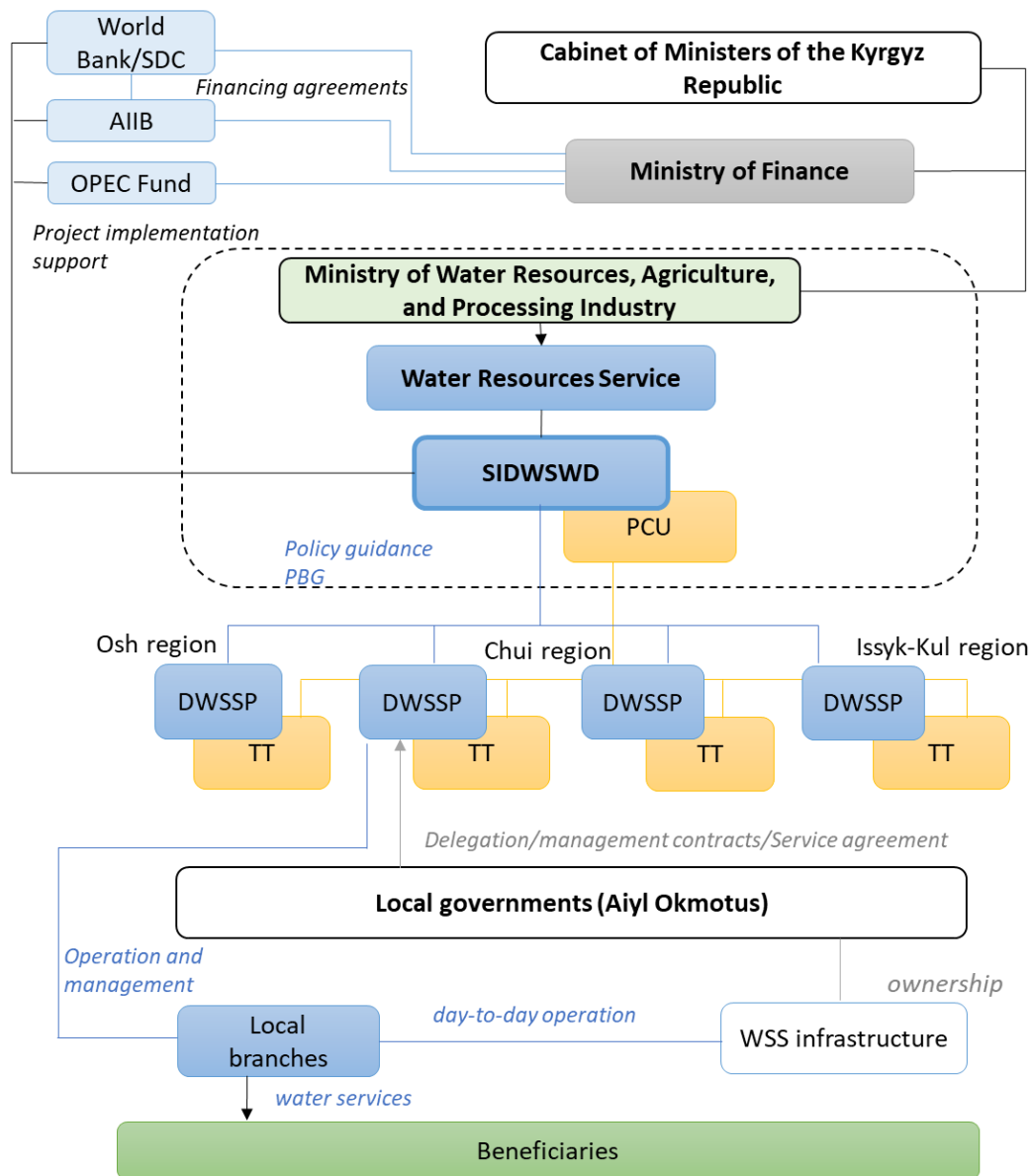
Kyrgyz Republic: Water Supply and Sanitation Universal Access Program

Implementation Arrangements

- 1. The institutional and implementation arrangements are built on the experience from previous projects and aligned with ongoing institutional reforms to strengthen the sustainability of service delivery.** Key institutions involved in the implementation include the MoF, the MWRAPI, WRS, and its SIDWSWD, DWSSPs, and AOs at the local level. The WRS acts as an executive agency, which provides strategic guidance and coordination of the Project, and ensures compliance with domestic procedures as may be required for the Project's implementation. The SIDWSWD, as an implementing agency, will lead the Project's implementation, including Program and Project management, and establish the required capacity at the regional and local level for service delivery, including implementing reforms and the PBGs to establish autonomous service providers and, where feasible, public-private partnerships. ADB and IsDB projects will still rely on ARIS for the implementation of the infrastructure components and SIDWSWD for the capacity building and PBG for the DWSSPs.
- 2. The existing PIU of the SIDWSWD responsible for the implementation of the ongoing CRWSP will be expanded and further strengthened to manage the proposed program.** The PIU will be restructured into a PCU at the central level based in Bishkek, with a dedicated team of specialists, comprising a Project coordinator for the PCU, and dedicated Project staff, including a manager, a financial manager, a procurement specialist, a disbursement specialist, an environmental specialist, a social development specialist, water supply and sanitation engineers, an institutional development specialist, a M&E specialist, a communications specialist, and administrative staff. The PCU will coordinate the management of the PBGs under the SIDWSWD oversight, including the selection of a verification agent.
- 3. At the district level, technical teams will be established within the DWSSPs to support Project implementation, and M&E.** The technical teams will include at least a district Project coordinator and water engineer, a sanitation officer, an environmental and social officer, and an institutional development officer. These technical teams will work closely with the PCU at the SIDWSWD and be responsible for day-to-day Project oversight, including technical support in the implementation of the PBGs and performance improvement plans. They will also participate in the technical development, review, and evaluation of relevant activities.
- 4. Two types of agreements will guide Project implementation.** The World Bank, AIIB, and OPEC Fund will sign financing agreements with the MoF as the Recipient of the credits. The World Bank will also sign a grant agreement with the MoF, as the grant administrator for SDC. SIDWSWD will sign performance agreements with each DWSSPs, with terms acceptable to the World Bank. All Project operational modalities will be detailed POM, which will be adopted prior to Project effectiveness. The implementation arrangements are presented in Figure 1.1.

Procurement

- 5. Applicable Procurement Framework.** Activities under the Project will be subject to the Bank's Procurement Framework. Procurement of all contracts will follow the procedures specified in the Bank's Procurement Regulations for IPF Recipients-Procurement in Investment Project Financing Goods, Works, Non-Consulting and Consulting Services, dated September 2023 (Procurement Regulations). The Project will also be subject to the World Bank's Anti-Corruption Guidelines, dated July 1, 2016. The procurement and contract management processes will be tracked through the STEP system.



TT = Technical teams at the local levels

Figure 1.1 – Project Implementation Arrangements

6. As required by the Procurement Regulations, the PSD is developed, based on which a PP is prepared, setting out the selection methods to be followed by the Borrower during Project implementation when procuring goods, works, and non-consulting and consulting services financed by the World Bank. The procurement approaches for key packages have been determined in the PSD as described in the following paragraphs.

7. **Procurement approach for key goods, works and non-consulting services contracts.** Civil Works and Goods will be procured through Request for Bids procurement methods with open international or open national market approach. All civil works packages following open international market approach will use relevant rated criteria.



8. **Procurement approach for key consultancy contracts:** The Project will finance audit services, which will be procured through the Least Cost Selection procurement method. Consultancy packages for the development of detailed engineering designs, author supervision, and technical supervision will be procured through Quality and Cost-based Selection methods. It was discussed and agreed that technical supervision will be provided by consulting firms, to be competitively selected.

9. **Delegation of Procurement Function under PBGs for DWSSPs.** The execution of procurement under **PBGs** will be delegated to DWSSPs. The grants may include expenses for the following categories: goods, services, minor civil works, training, and technical assistance. Procurement responsibilities under the PBG program will be delegated to the Grantee. The grantee will be responsible for carrying out the bidding process and concluding contracts for all packages under the PBG program. Procurement responsibilities, procedures, and sample documentation will be described in the POM and in the PBG Handbook. The procurement records for PBGs will be kept in PCU files, and a summary of the contract award information will be provided bi-annually to the Bank in the agreed format. Annual audit will include a requirement for procurement audit of the PBGs.

10. **Key conclusions from the conducted market analysis.** The current state of the construction market has sufficient competition to offer the best value under all equal conditions, even taking into account the regionality, so that the construction market is quite competitive. There is also a need for an adequate and safe system of transferring the facility to the contractor and vice versa with the participation of third parties, possibly various public associations and active social groups.

11. **Procurement risks analysis.** A procurement capacity assessment was performed by the World Bank using the Procurement Risk Assessment and Management System. Based on the assessment and taking into account the overall procurement environment in the country, the overall Project risk for procurement is assessed as Substantial. The main procurement risks are detailed in Table 1.1.

Table 1.1. Procurement Risks

• Procurement Planning: Fluctuations in gas and oil prices, as well as fuel costs, may impact the final product's price, potentially leading to increased cost estimates. • Procurement Process: The capacity of Project beneficiaries to prepare comprehensive designs and technical specifications for defined investments is limited. This limitation heightens the risk of undue influence on government officials overseeing procurement decisions, which could compromise the integrity and accountability of these decisions. • Potential Procurement Delays: Historical data indicates that procurement delays are likely, stemming from insufficient procurement capacity and market constraints. Additionally, depreciation of the local currency may deter potential bidders from submitting bids or quotations in that currency. • Potential Contract Implementation Delays: Present challenges in logistical services, including the transportation of goods, are adversely affecting the timely delivery of goods, particularly from the Commonwealth of Independent States and other European zone countries. • Overall Procurement Environment: The procurement environment is currently characterized by instability, frequent amendments to Procurement Policy and Legislation, and a high incidence of corruption, as indicated by Transparency International's assessments.
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12. To mitigate the risks, the following actions, listed in Table 1.2, have been identified:

Table 1.2. Preliminary Risk Mitigation Measures

• Procurement Framework: All procurement activities will be conducted in strict adherence to the World Bank's Procurement Procedures, including requisite prior or ex-post reviews. The World Bank's governance and anti-corruption safeguards, especially those related to transparency and disclosure, will be actively promoted and rigorously enforced.
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- **Procurement Support and Responsibilities:** The PCU, bolstered by the expertise of additional consultants, will assume responsibility for the preparation of procurement documents. The World Bank will provide comprehensive implementation support to ensure meticulous oversight.
- **Procurement Planning and Monitoring:** Meticulous procurement planning, current cost estimates, and precise scheduling are imperative, including the timely preparation of technical specifications or Terms of Reference (TOR) under the vigilant supervision and monitoring of the World Bank, particularly by the country office. Proactive engagement with the market and strategic business outreach are essential for critical procurement packages.
- **POM:** The POM must delineate explicit deadlines and timelines for each stage of the procurement process, encompassing the selection of consultants as well as the procurement of goods, works, and services, to preclude unnecessary delays during Project implementation.
- **Contract Management and Oversight:** An increased focus on contract management training is necessary, augmented by consistent physical inspections during the World Bank's supervision missions. Delivery terms will be scrupulously reviewed to circumvent transportation through conflict-affected zones and to explore alternative routes.
- **Anti-Corruption Measures:** The application of the World Bank's Anti-Corruption Guidelines will be stringently applied, with the World Bank staff ensuring close supervision. The disclosure and publication of information regarding beneficiary ownership will be mandated.

13. Use of National Procurement Procedures. In accordance with paragraph 5.3 of the Procurement Regulations, when approaching the national market, as agreed in the Procurement Plan, the country's own procurement procedures may be used. The provisions of the Public Procurement Law (PPL) were assessed by the World Bank as partially consistent with the World Bank Procurement Regulations Section V – Para 5.3 -5.6 National Procurement Procedures. When approaching the national market (as specified in the PP tables in STEP), the Borrower may use the procedures of “Unrestricted” and “Request for Quotations” methods as set forth in the PPL dated April 14, 2022, № 27 with amendments dated January 30, 2023 N19 and July 25, 2023 N147. National Procurement Procedures shall be subject to the provisions of paragraph 5.4 of the Procurement Regulations and the additional conditions specified in the PPSD and PP. To promote transparency, efficiency, and value for money under the country's public procurement system, the PPL provides for e-procurement system. The e-procurement system was assessed by the World Bank and the Project may use it for procurement of simple goods and minor works.

14. Selection of PCU personnel, Training and Operating Costs. The Project will finance operating costs for the PCU. When required, the PCU's personnel will be selected on the basis of experience, qualifications, and capability to carry out the assignment. The selection shall be carried out through the comparison of the relevant overall capacity of at least three qualified candidates among those who have expressed interest in the assignment. Detailed procedures will be outlined in the POM. The PCU will develop a detailed training plan and prepare an annual operational budget for the Bank team's review and clearance. Operating costs and training will be financed as per the annual budget approved by the Bank.

15. Record keeping. All records (except records under PBGs) pertaining to award of tenders, including bid notification, register pertaining to sale and receipt of bids, bid opening minutes, bid evaluation reports and all correspondence pertaining to bid evaluation, communication sent to/with the Bank in the process, bid securities, and approval of invitation/evaluation of bids would be retained by respective agencies and uploaded in the STEP.

16. Disclosure of procurement information. The following documents shall be disclosed: (a) PP and updates; (b) an invitation for bids for goods and works for all contracts; (c) request for expression of interest for selection/hiring of consulting services; and (d) contract awards for goods, works, non-consulting and consulting services.

17. The following details shall also be published in the United Nations Development Business and Bank's external website: (a) an invitation for bids for procurement of goods and works following open international market approaches; (b) Request for Expression of Interest for selection of consulting services following open international market approaches; and (c) contract award details of all procurement of goods and works and selection of consultants.



18. Fiduciary oversight by the Bank and procurement supervision. The Bank shall prior review contracts as per prior review thresholds set in the PPSP/PP. All contracts not covered under prior review by the Bank shall be subject to post-review during implementation support missions and/or special post-review missions, including missions by consultants hired by the Bank. Two half-yearly missions are envisaged for procurement support and supervision of the proposed Project.

Table 1.3. Summary Procurement Plan

No	Description	Total PP (US\$)
1	Goods	3,380,000
1	Works	165,574,389
2	Consulting Services	6,688,938
3	Grants Results-based grants to service providers	18,000,000
4	Trainings	630,000
5	Additional Project Costs (Operational Expenses) - PCU Staff and Non-Procured	5,646,960
6	Component 5 – Contingent Emergency Response	-
Total		199,920,286

Financial Management

19. Implementing Agency. FM including the flow of funds, budgeting, accounting, reporting, internal controls, and external audit, will be implemented by the SIDWSWD's PCU. An FM assessment was carried out to determine the FM implementation risk and FM arrangements at the IA, including accounting, reporting, planning, budgeting, and internal controls, and staffing. The FM arrangements were assessed to be overall *Satisfactory* to the Bank. The inherent risk of the Project after applying risk mitigation measures is rated as Moderate; the Control Risk and the overall residual FM risk are also considered to be *Moderate*.

20. Budgeting and planning. The existing PCU has acceptable planning and budgeting capacity in place. The Project's budget is being approved by the MoF. The Project's annual budget will be prepared by the Financial Manager based on the PP and working plan, in coordination with the PCU director, Component Coordinators and Procurement Specialist. The budget will be submitted for the MoF's approval with endorsement by the Minister of the MWRAP. Budget revisions that take place twice a year are also approved by the MoF. The budget data will be entered into the accounting software and used for monitoring of variances between actual and budgeted expenditures.

21. Accounting and Reporting. The cash basis of accounting will be applied to the Project's accounting. The IA will maintain its current accounting systems with the necessary customization to accommodate this Project. Project-management-oriented IFRs will be prepared under the Project. The PCU will produce a full set of IFRs every calendar quarter throughout the life of the project. The format of IFRs have been agreed upon before negotiations and include (i) Project Sources and Uses of Funds, (ii) Uses of Funds by Project Activities, (iii) Project Balance Sheet, (iv) DA Statement, and (v) Withdrawal Schedule. IFRs will be produced by the accounting software. These financial reports will be submitted to the Bank within 45 days of the end of each calendar quarter. The annual audited Project's financial statements and audit report, together with the management letter, will be provided to the Bank within six months of the end of each fiscal year as well as at the closing of the Project.

22. Internal Controls. The internal controls system at the existing PIU was assessed to be capable of providing timely information and reporting on the Project. The FM chapter of the POM is well prepared and fully documents accounting and financial reporting policies and procedures of existing Project such as internal control procedures, including



authorization of expenditures and approval of the payments, bank reconciliations, verification of expenditures eligibility by the Financial Managers; description of financial documents flow/circulation; indication of eligible cash transactions, budgeting procedures, formal reconciliation procedures of Project records with Client Connection, safeguards for assets, etc. Similar internal control systems will be maintained for the purpose of the Project. Expenditures incurred by the Project will be authorized by the Director of the IA for the respective components and verified for the eligibility and accuracy by the financial managers. Similarly, a POM will be prepared for this Project to reflect specific activities of the Project, including Chart of Accounts, Audit TOR, frequency of submission, format of IFRs, and so forth.

23. Staffing. The PIU has experienced FM staff, consisting of Financial Manager responsible for overall FM arrangements of the Project. The Financial manager has sufficient experience working in WB-financed projects. FM Manager will be in charge for the overall FM arrangements of the WASUAP as well. There is also an experienced disbursement specialist at the PIU. At a later stage, an additional disbursement specialist will be hired by the IA to evenly distribute the workload.

24. External Audit. The Project audit will be conducted (i) by independent private auditors acceptable to the Bank, on the Terms of Reference (ToR) acceptable to the Bank and selected by the PCU; and (ii) according to the ISA issued by the International Auditing and Assurance Standards Board of the International Federation of Accountants. The ToR will include (i) audits of financial statements, (ii) assessments of the accounting system, and (iii) a review of the internal control mechanisms. The audited financial statements will be disclosed to the public in a manner acceptable to the Bank. Following the Bank's formal receipt of these statements from the Borrower, the Bank makes them available to the public in accordance with the World Bank Policy on Access to Information.

25. Disbursements. Disbursements from the Credit Account will follow the transaction-based method, i.e., traditional Bank procedures including advances to designated accounts, direct payments, Special Commitments, and reimbursement (with full documentation and against Statements of Expenditures). A designated account for the credit funds will be opened by the PCU in the commercial bank acceptable to the World Bank. For payments above the minimum application size, as will be specified in the disbursement and financial information letter (DFIL), the PCU may submit withdrawal applications to the World Bank for payments to suppliers/contactors/consultants directly from the Credit Account, as applicable. Disbursement arrangements will be detailed in the DFIL. Details of cash flow arrangements will be described in the FM part of the POM.

26. The eight DWSSPs benefiting from the PBGs are likely to have limited FM capacity in the early stages of operation. The DWSSPs will be established with support from the Project, and in the early years of operation, the service providers' accounting systems may require significant improvements, including planning and budgeting, internal controls, funds flow, financial reporting, and auditing. The Project will provide technical support as part of the start-up package to establish the procedures and tools for an improved FM.

27. Funds transferred from SIDWSWD to the DWSSPs for service improvements under Component 3, will be liquidated quarterly. The liquidation report will be supported by appropriate documentation including bank reconciliations documenting actual costs incurred on eligible expenditures under the PBGs. Each DWSSP will open a separate local currency bank account and be subject to internal control policies, procedures, and audits in compliance with local legislation and World Bank-financed operations. Should the DWSSPs spend any funds on ineligible expenditure, they will not be able to access additional funds under the Project until all funds for ineligible expenditure are reimbursed and clearance is given by IDA to reinstate the flow of funds.

Implementation Support Plan

28. Technical Implementation Support. During Phase 1, the task team will continue to engage an experienced civil engineer to ensure the technical quality of outputs. Specifically, a civil engineer with a specialization in WSS will be engaged to review all designs prior to tender. Given that earlier Bank-financed projects in the Kyrgyz Republic have



experienced issues with sub-optimal designs and underestimated investment costs, and the generally low, albeit improving, capacity of the local consulting industry, reviewing designs is required to mitigate existing risks. During the construction phase, the engineer on the team will provide supervision support to ensure the quality of works and safety as well as to advance discussions on the O&M strategies by the recipient participating AO's. Given the shortcomings in construction supervision observed in earlier projects, particular attention will be placed to ensure that supervision systems are in place and are being followed closely. Among others, weekly planning documentation and daily supervision records will be reviewed regularly. Technical implementation missions will be implemented three times a year during the first 18 months of Project implementation, followed by bi-annual supervision missions. The engineer will carry out site visits where works are ongoing or where service has recently commenced.

29. Institutional Strengthening Support. WASUAP has a focus on strengthening the water sector institutions at the local and central level for improved service delivery. The Bank will offer technical support through this process by deploying specialized skill sets relevant to the planned reform activities. This will specifically include an Institutional Specialist, experienced in water sector reforms and to support the design and implementation of the PBGs, together with a Financial Specialist, with extensive experience in utility finances, tariff structuring, investment planning and financial regulation. Institutional strengthening support will be implemented two times a year for the full Project period.

30. Procurement Supervision and Ex-post Review. Routine procurement reviews and supervision will be provided by the Procurement Specialist based in the country office. In addition, two supervision missions are expected to take place per year during which ex-post reviews will be conducted for the contracts that are not subject to Bank prior review on a sample basis. One ex-post review report will be prepared per fiscal year, including findings of physical inspections for contracts awarded during the review period.

31. FM Implementation Support. During Project implementation, the Bank team will supervise the Project's FM arrangements in the following ways: (i) review the Project's quarterly IFRs as well as the Project's annual financial statements, the auditor's reports and management letters, and remedial actions recommended in the auditor's management letters, and (ii) during the Bank Team's on-site missions, review the following key areas: Project accounting and internal control systems; budgeting and financial planning arrangements; disbursement arrangements and financial flows, including counterpart funds, as applicable; and any incidences of corrupt practices involving Project resources. As required, a World Bank-accredited FM Specialist will participate in the implementation support and supervision process.

32. E&S Safeguards Implementation Support. A Bank Environmental Specialist will review the implementation of the Project's ESMF and provide guidance to the IA's Environmental Specialist to ensure compliance with the Bank's environmental safety guidelines. Similarly, a Bank Social Specialist will review the implementation of the Project's Resettlement Action Plans (if any) and provide guidance to the PCU to ensure compliance with the Bank's social safeguards guidelines.

Table 1.4. Skills mix required for the duration of Project implementation

Skills Needed	Number of Staff Weeks	Number of Trips	Comments
Task Team Leaders	30	10	Mixed: field and non-field-based staff
Water and Sanitation Engineer	20	10	Mixed: field and non-field-based staff
Institutional Specialist	10	5	Non-field based staff
Financial Specialist	10	5	Non-field based staff
Procurement Specialist	8	8	Field-based staff
FM Specialist	4	4	Field-based staff
Environmental Specialist	4	6	Field-based staff
Social Specialist	8	8	Field-based staff



ANNEX 2: Detailed Project Design, Economic and Financial Analysis, and Project Map

COUNTRY: Kyrgyz Republic

Kyrgyz Republic: Water Supply and Sanitation Universal Access Program

Introduction

1. **The Government of the Kyrgyz Republic has set an ambitious target to supply centralized drinking water and achieve universal access to at least basic water supply by 2030.** The government strategy builds upon more than two decades of infrastructure investments that have resulted in significant progress on WSS access and related health outcomes. Between 2001 and 2023, 839 of the 2,014 villages benefited from infrastructure investments, leading to an additional 1.7 million people with access to improved water supply and a cumulative 6.5 million people with access to basic water supply. There are still 363 villages (more than 378,000 people) without a water supply system. Of these, 84 (96,600 people) have planned interventions, and the remaining 279 villages require financing. An additional 617 settlements (rural and urban) require some upgrade or service expansion to cover approximately 280,000 people. The program aims to gradually address the infrastructure gap by covering the last mile of villages while supporting service upgrades in the served settlements. Figure 2.1 presents the coverage evolution through the multi-phased program. By the end of the program in 2035, it is expected that all settlements have access to a functioning water supply system.

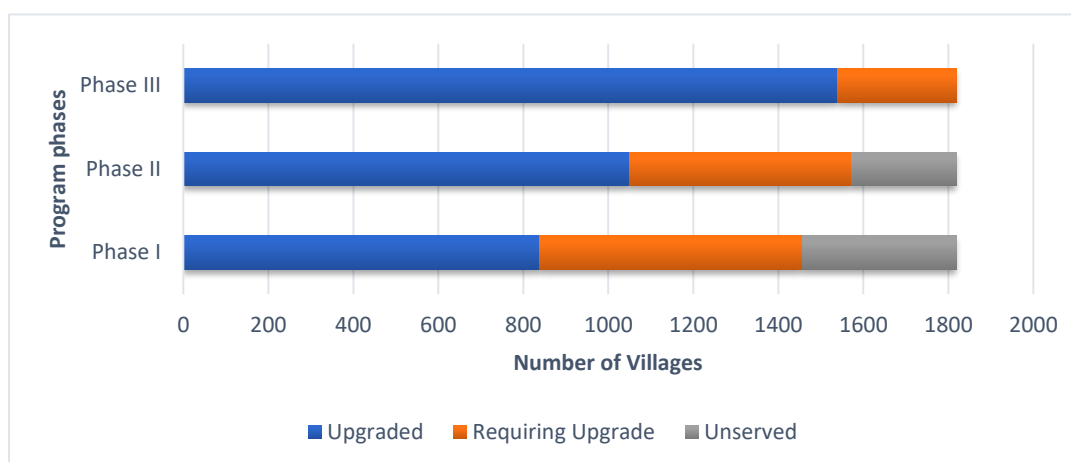


Figure 2.1 – Coverage evolution through the MPA

2. **The Project will particularly focus on service quality and sustainability.** Out of the 1,267 registered service providers (municipal enterprises, CDWUUs, informal service providers), only nine percent are effectively covering their operating costs. The remaining 91 percent can hardly cover some minor maintenance works (18 percent) or very limited operating costs (73 percent). The poor operation performance is a major burden to the sector and is leading to reinvestment in some systems. The Project will particularly focus on incentivizing the operational and financial performance of the water supply services.

Project Description

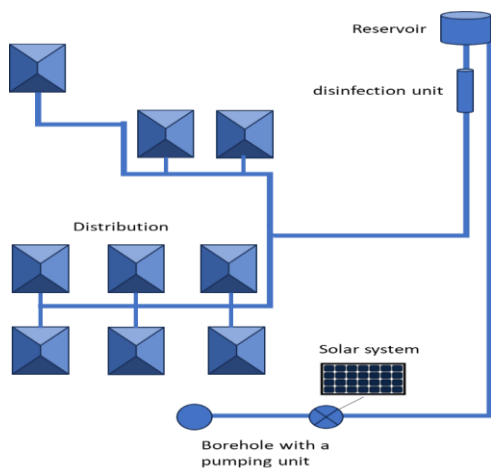
3. **Phase 1 will focus on three strategic areas:** (i) establish a framework for accelerating WSS infrastructure investments and capacity development in the country; (ii) progressively develop climate-resilient WSS infrastructure and enable water supply access in unserved settlements of the Kyrgyz Republic, and (ii) test and quick start the implementation of a PBG financing mechanism to incentivize service delivery efficiency and enable climate adaptation. Phase 1 will be implemented over a four-year period (2025–2029).



Component 1 – Water Supply Investments (US\$143.75 million, of which US\$61.55 million IDA credit, US\$8.45 million IDA shorter maturity loan, US\$50.00 million co-financing credit from the AIIB, US\$ 3.75 million parallel financing grant from AIIB and US\$20.00 million co-financing credit from the OPEC Fund)

4. This component will finance the construction and upgrade of water supply systems in 126 unserved rural villages in priority regions, targeting **450,000** people. World Bank and AIIB co-financing for this component will focus on priority investments in Chui, Issyk-Kul, and Osh regions. The Project will prioritize the use of gravity systems, energy-efficient pumping equipment, and solar energy for new water supply schemes (which is around 38percent of the systems) to minimize GHG emissions. Water source development will target groundwater exploitation (in at least 80 percent of the systems – Figure 2.2) to minimize the impacts of climate change, and network construction will utilize climate-resilient infrastructure design such as materials resistant to high-temperature variations and withstand ground movements in landslide-prone areas in Osh region. All water intakes will include source protection and reforestation interventions to prevent contamination and enable water conservation and aquifer recharge. The system design will adopt the WB-RWIDB to minimize the risk of water supply disruption due to landslides and other climate-related events and will follow a participatory approach, with consultations involving disabled people, building on local knowledge and historical data. Component 1 also includes an allocation for DEDs financed through AIIB grant (US\$3.5 million), construction supervision, household connections up to the meter, and contingencies to address potential unanticipated technical challenges that would hinder the achievement of water supply objectives in the Project areas. Additional village water supply system upgrades to ensure universality and service improvements will be covered through **ADB** financing in **the Naryn Region⁵³ (US\$ 32 million serving 36,000 people) and IsDB financing in the Jalalabad Region.**

Scheme with a groundwater source



Scheme with a spring source

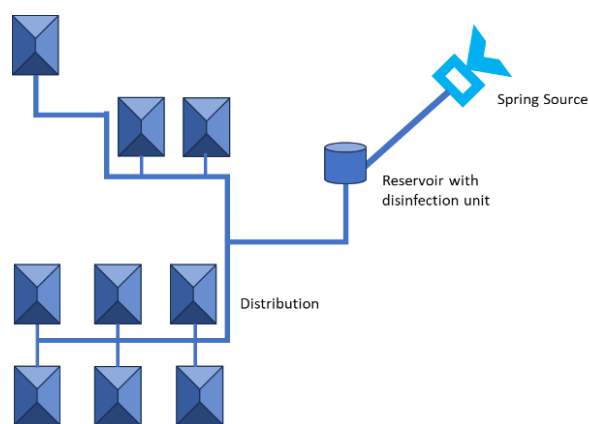


Figure 2.2 – Schematic representation of rural water systems

5. Subcomponent 1.1 – Water Supply Access to the Unserved (US\$43.70 million). This sub-component will finance the design and construction of climate-resilient water supply systems in 48 currently unserved settlements in the target regions. The first phase will prioritize settlements with a population of at least 1,000 people to ensure the economic

⁵³ The AF for the Kyrgyz Republic: Naryn Rural Water Supply and Sanitation Development Program was approved in September 2024 and comprises a \$27 million concessional loan and a US\$5.35 million grant from the Asian Development Fund, and a contribution from the Government of the Kyrgyz Republic of \$6.45 million. Using a results-based approach, the AF will help to scale up interventions in centrally located Naryn region—raising the initial target of 64,000 people reached to 100,000. The funding also enables an increase in the number of education and health facilities that have separate toilets for women and men from 21 to 37. Source: <https://www.adb.org/news/additional-adb-financing-expand-water-supply-sanitation-coverage-kyrgyz-republic>



viability of the water supply services. Specifically, the Project will finance resilient water source development, water treatment infrastructure (mostly ultraviolet disinfection systems) to ensure high drinking water quality, robust transmission and distribution networks capable of withstanding extreme weather conditions targeting **128,000** people in Chui, Osh, and Issyk-Kul regions. All water connections will be metered to ensure sustainable water use.

6. Subcomponent 1.2 – Water Supply Upgrades (US\$100.05 million). This sub-component will finance the engineering design, rehabilitation, and upgrade of targeted water supply system upgrades in 78 villages and district centers in Aksu, Alai, Aravan, Karakulja, Kara Suu, Panfilov, Sokuluk, and Uzgen districts (around **322,000** beneficiaries). The upgrades will prioritize investments with demonstrated impacts on service reliability, financial viability, and climate resilience in (i) settlements with low access and dilapidated infrastructure, (ii) economically important settlements (including secondary cities and district centers) with poor performing infrastructure and low service quality, and (iii) other settlements requiring service upgrades. Specifically, the Project will finance investments focused on source and storage increase and management efficiency, network hydraulic improvements to reduce non-revenue water and expansion to secure water supply to communities affected by droughts in the face of climate variability, ensure efficient water management, and reduce vulnerability to climate impacts. The water supply upgrades are aligned with the country's climate resilience goals, ensuring reliable access to water services amidst changing climate conditions. Ten of the larger systems serving small towns (approximately 145,000 people) will also benefit from solar power installations and pumping station upgrades to improve energy efficiency. The summary of the proposed water supply investments and estimated costs is presented on Table 2.1. Cost estimates took into account future population growth and contingencies to cover for market changes.

Table 2.1. Cost estimates for water supply investments

No	Rayon	Village	Population 2024	Number of population 2028	Population 2035	Construction costs (US\$)	Design and Supervision costs (US\$)	Total costs (US\$)
Total for the country		126	426,164	450,357	496,047	136,502,711	6,825,136	143,327,847
Sub-component 1.1		48	120,946	127,812	140,779	41,378,346	2,068,917	43,447,264
<i>Osh oblast</i>		47	119,511	126,296	139,109	40,760,330	2,038,017	42,798,347
	Karakuljinskiy	10	29,120	30,773	33,895	9,599,955	479,998	10,079,953
	Kara Suu	19	41,297	43,641	48,069	14,587,677	729,384	15,317,061
	Uzgen	8	14,484	15,306	16,859	5,437,834	271,892	5,709,726
	Aravan	7	17,884	18,899	20,817	5,903,811	295,191	6,199,002
	Alai	3	16,726	17,676	19,469	5,231,053	261,553	5,492,605
<i>Issik Kul Oblast</i>		1	1,435	1,516	1,670	618,016	30,901	648,917
	Aksuu	1	1,435	1,516	1,670	618,016	30,901	648,917
Sub-component 1.2		78	305,218	322,545	355,268	95,124,365	4,756,218	99,880,583
<i>Osh oblast</i>		35	122,013	128,940	142,021	37,664,719	1,883,236	39,547,955
	Karakuljinskiy	1	4,535	4,792	5,279	1,319,665	65,983	1,385,648
	Kara Suu	3	10,541	11,139	12,270	3,203,046	160,152	3,363,199
	Uzgen	1	5,105	5,395	5,942	1,485,532	74,277	1,559,809
	Aravan	29	100,230	105,920	116,666	30,966,537	1,548,327	32,514,864
	Alai	1	1,602	1,693	1,865	689,939	34,497	724,436
<i>Chui Oblast</i>		29	147,703	156,088	171,924	45,430,041	2,271,502	47,701,543
	Sokuluk	23	123,611	130,628	143,881	37,778,440	1,888,922	39,667,362
	Panfilov	6	24,092	25,460	28,043	7,651,600	382,580	8,034,180
<i>Issik Kul Oblast</i>		14	35,502	37,517	41,324	12,029,605	601,480	12,631,085
	Aksuu	14	35,502	37,517	41,324	12,029,605	601,480	12,631,085

Component 2 – Sanitation Development (US\$29.00 million IDA credit)

7. This component will finance priority on-site sanitation improvements focused on the containment element of the service chain, FSM services, and TA to design more complex sewerage improvements. The cost estimates for sanitation interventions are presented in Table 2.2. The Project will finance: (i) construction, rehabilitation/retrofitting of decentralized sanitation systems, including toilet upgrades for vulnerable households and selected public institutions (schools, kindergartens, hospitals, and health centers) in all 126 target settlements, thereby enhancing community resilience to climate-related health risks; (ii) TA, equipment and works for FSM services improvement, including the



guidelines for fecal sludge collection, transport, treatment, and re-use, which contributes to climate mitigation by preventing the release of methane from untreated waste; (iii) the implementation of a sanitation marketing and behavioral change strategy to incentivize the adoption and use of improved WASH and behaviors that are essential for climate resilience and mitigating the pressures on water resources; and (iv) TA for the assessment and engineering design of centralized and decentralized sewerage infrastructure in priority settlements where technically, financially, and environmentally feasible, aiming to improve water quality and public health while reducing environmental pollution and enhancing ecosystem resilience against climate impacts. A total of 3,300 vulnerable households will benefit from the toilet upgrades, 132 schools, 132 kindergartens, and 125 health centers will benefit from WASH improvements.

8. The household toilet construction will be implemented through output-based contract with contractors. The output-based contract will incorporate the demand mobilization in the contractor's contract. The Project will engage sanitation marketing TA to provide strategic design and lead the implementation of the marketing campaign targeting the beneficiary households. The sanitation marketing specialists will also provide TA, materials and training to contractors, and the district technical teams to engage in field level marketing. The contractor's marketing costs will be reflected in the bid price. In addition, the PCU will make use of its existing social capital and community level structures to support the marketing effort. Interested contractors will bid on a unit cost basis for each standard toilet. Household contribution will be determined based on the socioeconomic conditions and will be paid upfront (upon 'signing up' households) through the digital payment mechanism to be established by the PCU for toilet applications. Payments will be structured in a way that a proportion of the price of each unit is paid to the contractor upon verification of household contributions, and the other proportion is paid upon verified completion of construction of the unit per pre-specified quality standards.

9. The Project will finance FSM business development support to private and municipal operators on a competitive basis. Operators will be selected based on their service improvement plan, including (i) quality of services at an affordable and predictable price, (ii) willingness to be regulated and join a service monitoring platform, and (iii) expanding emptying services in currently unserved areas. The Project will specifically finance market assessment TA, business structuring, pricing, and customer services, and the development of call centers and other IT platforms to facilitate FSM service delivery. Further, the Project will finance TA for the development of guidelines for fecal sludge collection, transport, treatment, and re-use, with specific procedures for the development of bio-products and options for preventing the release of methane from untreated waste.

Table 2.2. Cost estimates for sanitation investments

No	Rayon	# of Schools and Kindergarten	School toilets (US\$)	Kindergarten toilets (US\$)	# of clinics	Clinics toilets (US\$)	Total cost for on-site sanitation systems, USD	Design and supervision (US\$)	Total cost for institutional sanitation (US\$)	# of Vacuum trucks	Cost for FSM, including equipment (US\$)	Cost of hh sanitation (US\$)	Total cost (US\$)
Total for the country		132	9,170,000	3,960,000	125	5,000,000	18,130,000	906,500	19,036,500	38	1,890,000	8,241,678	29,168,178
Osh oblast		83	5,740,000	2,490,000	81	3,240,000	11,470,000	573,500	12,043,500	25	1,230,000	4,577,421	17,850,921
	Karakuljinskiy	10	700,000	300,000	10	400,000	1,400,000	70,000	1,470,000	3	165,000	535,795	2,170,795
	Kara Suu	22	1,540,000	660,000	22	880,000	3,080,000	154,000	3,234,000	7	330,000	885,149	4,449,149
	Uzgen	9	630,000	270,000	9	360,000	1,260,000	63,000	1,323,000	3	135,000	307,408	1,765,408
	Aravan	38	2,660,000	1,140,000	36	1,440,000	5,240,000	262,000	5,502,000	11	540,000	2,661,304	8,703,304
	Alai	4	210,000	120,000	4	160,000	490,000	24,500	514,500	1	60,000	187,765	762,265
Chui Oblast		33	2,310,000	990,000	29	1,160,000	4,460,000	223,000	4,683,000	9	435,000	2,633,913	7,751,913
	Sokuluk	25	1,750,000	750,000	23	920,000	3,420,000	171,000	3,591,000	7	345,000	2,128,736	6,064,736
	Panfilov	8	560,000	240,000	6	240,000	1,040,000	52,000	1,092,000	2	90,000	505,177	1,687,177
Issik Kul Oblast		16	1,120,000	480,000	15	600,000	2,200,000	110,000	2,310,000	5	225,000	1,030,343	3,565,343
	Aksuu	16	1,120,000	480,000	15	600,000	2,200,000	110,000	2,310,000	5	225,000	1,030,343	3,565,343

10. TA for the design of sewerage improvements. The Project will prioritize the design of sewerage pipelines capable of withstanding harsh weather conditions and wastewater and fecal sludge reuse solutions that enable methane capture for



settlements with multi-story buildings, whose sanitation systems are currently a source of environmental pollution, including groundwater contamination. This TA is also expected to leverage innovative and sustainable solutions for waste treatment and contribute to climate mitigation through improved wastewater management practices and adaptation by preserving the ecosystems in the receiving water bodies and the livelihoods dependent on it.

11. Component 3—Performance-based Service Improvement Program (US\$22.00 million, of which US\$13.00 million IDA credit and US\$9.00 million SDC co-financing grant) aims to enhance institutional and service delivery capacity. The Project will finance PBGs to DWSSPs to improve service delivery and sustainability, and the implementation of the professional and vocational training program. The PBG allocation per district was estimated using the WSS district aggregation financial model, which considered planned infrastructure improvements, enhanced service levels, and operational efficiency. Additional advisory TA will be provided through World Bank-executed trust fund financed by SDC.

12. Sub-component 3.1 – Service Delivery Improvements (US\$18.00 million) will finance priority WSS service improvement activities through PBGs to be managed centrally by the SIDWSWD. The Project will finance PBGs for eight DWSSPs (out of the ten planned for Phase 1) and will cover the start-up activities and operational and structural investments that are critical to enabling the operational and financial efficiency of the DWSSPs, and the resilience of water supply systems against climate variability. Funding for the PBGs will be aligned with the DWSSPs institutional and service development plans and will focus on increasing DWSSP's service management capacity, innovative solutions for service expansion and climate resilience, asset management and operations, digital solutions for customer services and feedback, tariff-setting procedures, financial management, and service monitoring and reporting. Additional training on utility management and rural services management will be provided with support from the SDC and Swiss experts.

13. The PBG will be structured in two parts: (i) a fixed part linked to the achievement of certain institutional prerequisites (such as the legal establishment of the DWSSPs, development, approval, and adoption of a service improvement plan, a gender improvement plan and, an approved tariff review framework and an auditable financial management system) to encourage the adoption of sustainable and equitable water management practices that are crucial for ensuring water services access to all, including vulnerable populations, thus contributing to climate adaptation. To incentivize efficiency in the establishment process, the fixed grant will only be available during the first 18 months after Project effectiveness. The amount for each DWSSP has been estimated based on the cost of the start-up equipment and other assets required by the DWSSP in the first two years of operation; (ii) a variable part linked to service delivery performance, including coverage, hours of supply, customer satisfaction, cost recovery, and NRW reduction, aiming to improve water use efficiency, thus supporting climate mitigation efforts by optimizing resource use.

14. The exact set of performance indicators will be defined in the performance agreement between the SIDWSWD and the DWSSPs. The variable part of the grant will be based on an analysis of the cost structure of water services in each DWSSP and revenue generation potential. The variable grant will be provided based on verified revenues collected through the tariffs, provided the DWSSP meets the minimum performance criteria on operational and customer services. The performance agreement will specify the maximum grant amount for each year within the Project period. This two-part design will enable service quality improvements to occur rapidly whilst the process of establishment and increasing revenue can occur gradually. Detailed estimates will be conducted prior to the signing of the performance agreements with each DWSSP, following a detailed financial analysis.

15. A Performance scorecard will be developed and agreed upon with each participating DWSSP. The scorecard consists of institutional prerequisites necessary for sustainable service provision and service and financial performance indicators. The first set of indicators includes a set of institutional actions that the DWSSPs, with support from the SIDWSWD, need to undertake within the first 18 months from Project effectiveness to establish the institutional framework for service provision. The second set includes indicators to track service provision performance. Considering the lack of a structured system for the collection and monitoring of operational and financial data on DWSSPs, the baseline data on service and financial performance indicators will only be available after the second year of the Project when the SIASAR has been



upgraded. SIDWSWD will agree on performance targets with each DWSSP and include them in the performance agreement, considering the baseline conditions.

16. Verification mechanism. The PBG will be subject to technical and financial verification to confirm the achievement of indicators on the performance scorecard. Acceptable verification will trigger the transfer of the grant to the DWSSP account. The minimum performance requirements on the scorecard will be defined in the performance grant agreement. The grant amount to be provided to the DWSSP in a specific period will be based on the financial analysis that demonstrates the gap to cover the service improvement costs. DWSSPs will only be permitted to fail the verification on three occasions. In the unlikely event that they fail on all three occasions, their participation in the system will be reconsidered.

Table 2.3 Estimated grant allocation for the beneficiary DWSSPs

Region	District		start up equipment (US\$)	Y2	Y3	Y4	Y5	Total (US\$)
Issyk Kul	Aksu	costs	1,202,500	354,375	311,850	343,035	377,339	1,947,477
		revenue		126,640	146,902	170,407	197,672	
		PBG	1,202,500	227,735	164,948	172,628	179,667	
Chui	Sokoluk	costs	1,520,000	472,500	415,800	457,380	503,118	2,300,554
		revenue		210,845	244,580	283,713	329,107	
		PBG	1,520,000	261,655	171,220	173,667	174,011	
	Panfilov	costs	1,060,000	450,000	408,375	449,213	494,134	2,469,034
		revenue		77,507	89,908	104,293	120,980	
		PBG	1,060,000	372,493	318,467	344,920	373,154	
	Aravan	costs	2,205,000	614,250	675,675	743,243	817,567	3,274,976
		revenue		351,477	407,714	472,948	548,620	
		PBG	2,205,000	262,773	267,961	270,295	268,947	
Osh	Karakuljinski	costs	1,417,500	529,740	582,714	640,985	705,084	2,531,848
		revenue		265,307	307,756	356,997	414,116	
		PBG	1,417,500	264,433	274,958	283,989	290,968	
	Kara Suu	costs	2,010,000	1,314,540	1,445,994	1,590,593	1,749,653	1,783,429
		revenue		1,248,861	1,448,679	1,680,468	1,949,343	
		PBG	2,010,000	65,679	(2,685)	(89,874)	(199,690)	
	Uzgen	costs	1,545,000	1,010,430	1,111,473	1,222,620	1,344,882	2,368,885
		revenue		762,957	885,031	1,026,635	1,190,897	
		PBG	1,545,000	247,473	226,442	195,985	153,985	
	Alai	costs	1,220,625	304,110	334,521	367,973	404,770	1,390,114
		revenue		245,117	284,336	329,830	382,603	
		PBG	1,220,625	58,993	50,185	38,143	22,168	
Total								18,066,317

17. Sub-Component 3.2 - Professional and Vocational Development Program (US\$4.00 million). The Project will finance the establishment of a national training center for water supply and sanitation hosted at the SIDWSWD, with a dedicated certificate-oriented, long-term capacity development program in response to the need to create a pipeline of water sector professionals with the skills to plan and manage efficient and sustainable water services. Training services will be selected on a competitive basis, taking into account the technical robustness of the proposed training program, demonstrated training methods, and effectiveness of the M&E systems. This program will also integrate lessons from the SDC-financed vocational training program targeting the youth in Kyrgyzstan.

18. The program will combine long-term studies with targeted short-term training on specific operational capacity gaps as identified by the DWSSPs. Other sector players, such as professionals working with consulting firms and contractors,



will also be eligible to attend the program training to enhance the capacity of service providers, particularly on non-conventional engineering topics such as FSM and sanitation marketing, among others. The training will also cover topics related to climate resilience adaptation, including the selection of construction materials and the design of robust WSS infrastructure to withstand climate variability and extreme events. The PVDP will also carry out extensive outreach and facilitate the enrollment of female participants in training programs designed to attract the youth and women to the water sector and enhance their technical and leadership skills. Training services to support the center will be selected on a competitive basis, considering the technical robustness of the proposed training program, demonstrated training methods, and effectiveness of the monitoring and evaluation (M&E) systems that demonstrate the training impact.

19. Component 4 – MPA Program Structuring and Management, and Institutional Development Support (US\$9.25 million, of which US\$9.00 million IDA credit, US\$0.25 million parallel grant from AIIB, and in kind counterpart financing) will finance technical assistance and institutional capacity building for the establishment of a program management framework and WSS infrastructure development capacity. Specifically, the component will finance (i) TA and training for the development of the program management tools, including standard E&S instruments, implementation manuals and protocols, to enhance coordination and Program implementation efficiency, and a M&E, and the M&E systems, including program results tracking and sector coordination; (ii) TA for the preparation of investment packages, engineering design, and E&S instruments for future interventions planned for the subsequent phases of the MPA, using WB-RWIDB; (iii) Incremental operating costs, TA and equipment for program management; (iv) TA, equipment and services to support a knowledge development program and communications strategy for the MPA aiming to build awareness and understanding among stakeholders about the importance of accelerated WSS access, service delivery sustainability and climate resilience; (v) TA for the preparation of service contract agreements between the operator and asset owner and the PBG independent verification; (vi) TA for targeted policy regulations, particularly focused on the implementation of the new service aggregation model and tariff framework.

20. Component 5 – Contingent Emergency Response (CERC - US\$0.00). This component will provide preparedness and rapid response measures to address disaster, emergency and/or catastrophic events in accordance with the applicable CERC Manual. Following an eligible crisis or emergency event, the Borrower may request the World Bank to reallocate undisbursed Project funds to support emergency response. This component would draw from the undisbursed financing resources under the Project from other Project components to cover eligible crisis or emergency, as needed.

21. The proposed project implementation plan is presented in Table 2.4, below. Water supply investments are divided in two batches, starting with the low-complexity and easy to design subprojects.



Table 2.4. Project Implementation Plan

Activity	Duration	Phase 1																								Phase 2																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																										
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		Months	J	F	M	A	M	J	J	A	S	O	N	D	J	F	M	A	M	J	J	A	S	O	N	D	J	F	M	A	M	J	J	A	S	O	N	D	J	F	M	A	M	J	J	A	S	O	N	D	J	F	M	A	M	J	J	A	S	O	N	D																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																						
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Economic and Financial Analysis

22. **The economic benefits from the Project will derive from improvements in the quality of water supply services to households and from sanitation services to households and schools, pre-schools and other public buildings.** Investments in rural water supply systems are expected to result in improved water quality, expanded access to improved water sources, increased duration of water supply, and reduced seasonal variation. Improving these services will enhance welfare by reducing coping costs (for example, time saved from water collection and reduced need for in-house drinking-water treatment, **reduced** household investments in water storage systems and, in some areas, elimination of the need to purchase water from tanker trucks). Improvements in the quality of water supply and sanitation services and practices through the WASH educational program are also expected to have positive welfare effects through improved health.

23. **While technically around 45 percent of households in the Project area do not have in-house or in-yard water connections, in fact the systems that exist are so dilapidated that there is effectively no water service, and households spend an average of up to 2 hours per day collecting water.** Many households still rely on unprotected standing water sources such as rivers, irrigation canals and shallow wells. And some villages need to purchase water from tanker trucks. As a result, households have significant coping costs. A household survey will be conducted by the PCU to provide baseline conditions for the villages targeted under Phase 1. An endline survey will be conducted prior to Project closing to assess end-of Project conditions and benefits.

24. **World Bank financing for the first phase of the program will invest in improved water supply services for approximately 450,000 residents in 126 villages, as well as provide improved sanitation facilities for households and public institutions.** The beneficiary population is projected to increase approximately 1.4 percent per year over the next 10 years, to 489,247; after 2035, this analysis has assumed a stable beneficiary population.

25. **The Project's economic analysis consists of a cost-benefit analysis projected over a 25-year period, including a four-and-a-half-year implementation period.** Costs consist of all investment costs of Components 1 and 3 (water supply investments and performance-based grants for water and sanitation service providers) and Component 2 (sanitation improvements for households and public institutions), as well as expected operating costs (assumed to be five percent of capital investment costs). Benefits included in the analysis include the value of decreases in the time spent collecting water and welfare gains at the household level associated with reduced need for in-house treatment (i.e., boiling of water) and reduced incidence of water-related diseases such as infectious hepatitis and diarrhea as a result of improved access to quality water and decline in the reliance on standing water sources. Expected benefits are based on results observed from earlier projects in the country, namely the Rural Water Supply and Sanitation Projects (RWSSP 1&2). Non-quantified benefits include reductions in household purchases of bottled water, water from tanker trucks, and investments in water storage assets, as well as the improvement in environmental quality resulting from improved sewage management and long-term benefits from improved school attendance by girls resulting from improved sanitation facilities.

26. **Operating costs and Project benefits are expected to be realized in the year following the implementation of each sub-project, and therefore increase incrementally each year until all Project investments are completed.** Given the rolling nature of sub-project construction, some benefits will begin to be observed in 2028, with full implementation (and benefits) in 2031.

27. Project benefits and the main assumptions of the analysis are as follows:

- **Project Investment Costs:**
 - Component 1: US\$ 143.75 million
 - Component 2: US\$ 29.0 million
 - Component 3: US\$ 22.0 million
- **VAT:** 12 percent, applied to 95 percent of costs of Components 1 and 2



- **Operating Costs:** Five percent of investment costs of Components 1, 2 and 3
- **Reduced time in collecting water:** Based on results obtained from earlier operations in the country (SRWSSDP, RWSSP 1&2), a reduction from approximately two hours to around 30 minutes of time for collecting water has been assumed. A rural average hourly wage of US\$0.74⁵⁴ is used as a conservative measurement of the opportunity cost of this time savings. Under these assumptions, the Project is projected to generate annual benefits of US\$39.0 million upon full Project implementation.
- **Reduced coping costs from boiling water:** It is common practice in the country for households to boil water prior to consuming it. Energy savings associated with reduced need to boil water due to improved water quality are estimated based on a 5-liters-per-day per-capita benchmark. We estimate that by the end of the Project only a negligible percent of households will boil water. Well-established benchmarks for the region are used in terms of energy requirements to boil water (0.09 kWh per liter of water), at an economic cost of only US\$ 0.03 per kWh.⁵⁵ Under these assumptions, the Project is expected to generate annual benefits of US\$1.57 million from energy savings linked to boiling water.
- **Benefits from reduced incidence of water-borne diseases.** Data on the incidence of water-borne diseases was previously collected in beneficiary villages under the SRWSSDP. With respect to hepatitis, we assume that people lose around 10 days of work/school and that treatment costs around US\$139 per incidence. Although underreported, diarrhea is significantly more common, with 15 percent of the population affected each year, although with lower lost productivity (one day) and treatment costs of only US\$3 per incidence. We find that a reduction in the above water-borne diseases could lead to annual savings of around US\$793,500 in reduced costs for treatment and US\$631,000 in reduced losses of days of work.

28. Based on the above assumptions, an implementation period of four-and-a-half years, and a total projection period of 25 years (including implementation), the projected net cost-benefit stream yields an ERR of 22.7 percent and an NPV, using a five percent social discount rate of US\$263.9 million.

29. **Estimates of GHG reductions have been carried out.** These indicate that the Project is expected to generate net reductions in GHG emissions of approximately 71,904 tCO₂-eq (a 47 percent decrease in emissions from the baseline scenario), through reduced energy use in the new and rehabilitated systems and from the reduction in the amount of water boiled at home (Subcomponents 1.1 and 1.2), for an average annual reduction of 2,876 tCO₂-eq over the 20 full years of post-implementation projections. Specifically, baseline emissions were estimated to be 1,218 tCO₂e/year from water boiling and 3,113 tCO₂e/year from the water supply systems and trucking. While Project emissions were estimated to be 1728 tCO₂e/year from rehabilitated systems and 1637 tCO₂e/year for new systems; energy saved from solar was estimated to be 201 tCO₂e/year, and carbon sequestration from reforestation was estimated to reduce CO₂e by 374 tCO₂e/year. In summary the emissions reduction was 115 percent below baseline for Component 1.1 (direct emissions reductions, carbon sequestration via reforestation, and the use of solar energy), while for Component 1.2, the reduction was 59 percent below baseline. Incorporating the value of the reductions into the economic analysis results in an ERR of 22.8 percent using the low value of GHG emissions and an ERR of 22.8 percent using the high value.

⁵⁴ Based on reported wage of a teacher in rural areas of KGS 24,000 per month, discounted by 50 percent.

⁵⁵ World Bank staff reported that both the economic cost and actual cost of electricity in rural areas is KGS 3 per kWh, or US\$ 0.034 per kWh. This is a very low number, and only a fraction of the economic cost presented in the World Bank's 2012 report: *Europe and Central Asia Balancing Act; Cutting Subsidies, Protecting Affordability, and Investing in the Energy Sector in Eastern Europe and Central Asia Region*. Because of this low cost of electricity, the benefits accruing from energy savings is lower than expected.



Project Map

